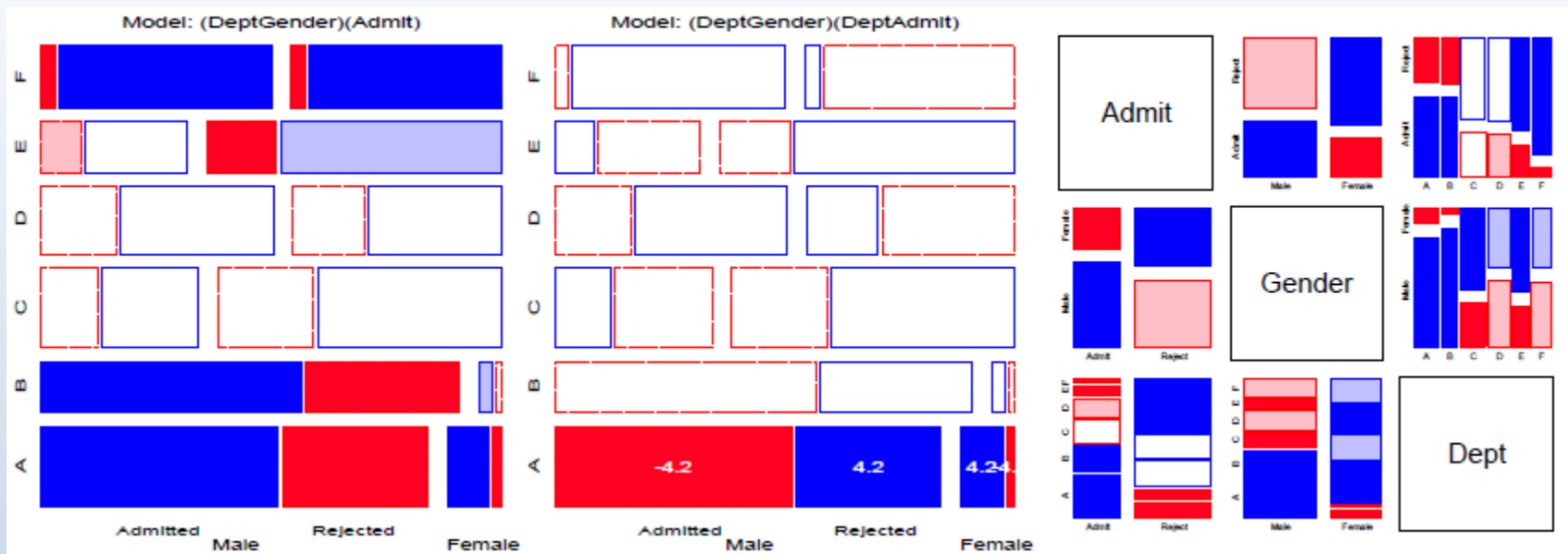
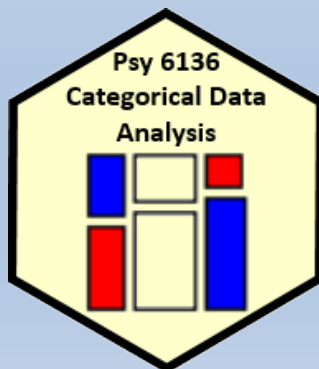


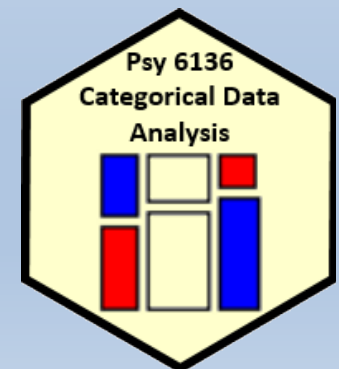


Loglinear models & mosaic displays



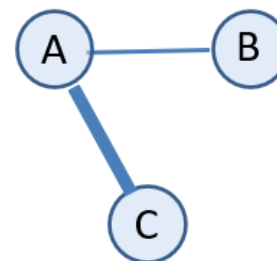
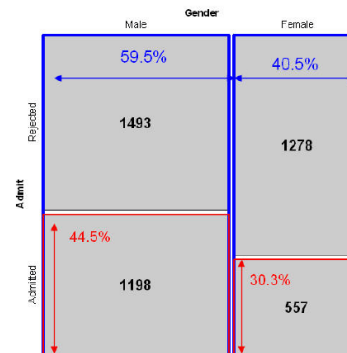
Michael Friendly
Psych 6136

<https://friendly.github.io/psy6136>



Today's topics

- Mosaic displays: basic ideas
- Models for count data
 - Fitting loglinear models
- Two-way tables
- Fitting & graphing LLMs
- Three-way tables: different kinds of independence
- Sequential plots & models
- Marginal & partial displays



Mosaic displays

- Similar to sieve plot, tile plot, using $\text{area} \sim \text{frequency}$
- Mosaic plots generalize more readily to n -way tables (subject to resolution of the display)
- Intimately connected to loglinear & generalized linear models
 - Can fit sequential models as variables are entered
 - Show the pattern of association not accounted for in a given model

Mosaic displays: basic ideas

Mosaic displays theory: Hartigan & Kleiner (1981); Friendly (1994, 1999)

UCB Admissions: Gender frequencies

Area proportional display for an n-way table

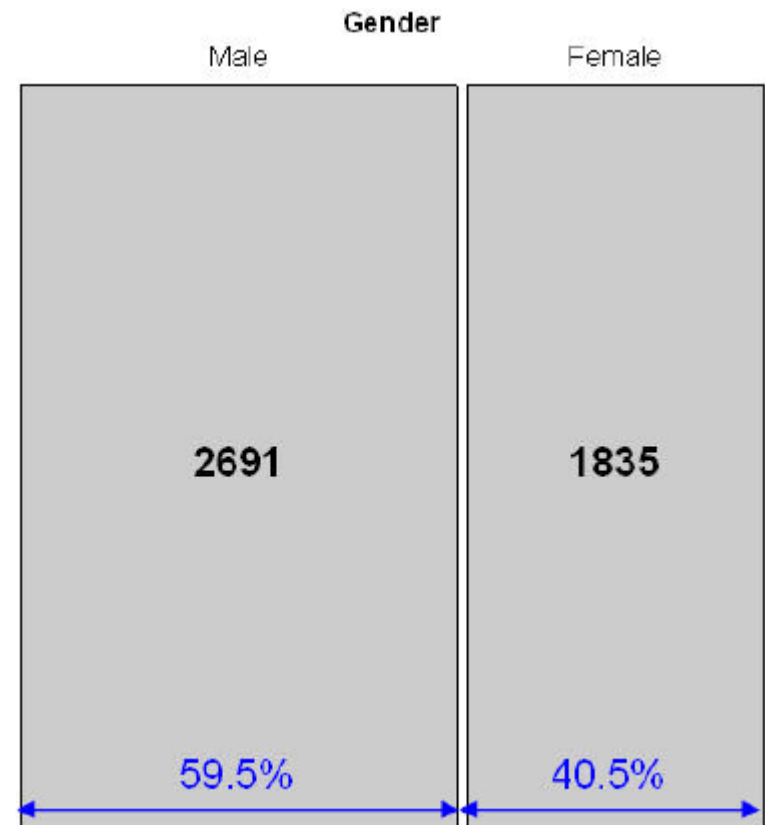
Tiles: **recursive** splits of a unit square, alternating H, V

V_1 : width $\sim$ marginal frequencies, n_{i++}

V_2 : height $\sim$ cond freq: $V_2 \mid V_1 = n_{ij} / n_{i++}$

V_3 : width $\sim$ cond freq: $V_3 \mid V_1, V_2 = n_{ijk} / n_{ij+}$

$\rightarrow$ Area $\sim$ cell frequency, n_{ijk}



Mosaic displays: basic ideas

UCB Admissions: Gender x Admit

Area proportional display for an n-way table

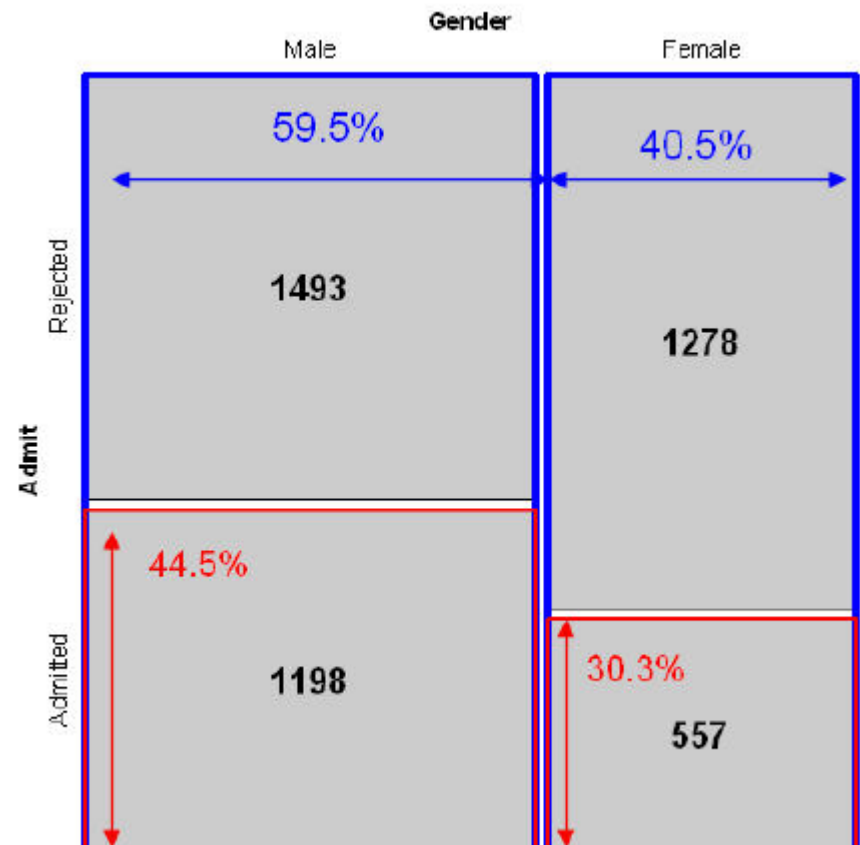
Tiles: recursive splits of a unit square, alternating H, V

V_1 : width $\sim$ marginal frequencies, n_{i++}

V_2 : height $\sim$ cond freq: $V_2 | V_1 = n_{ij} / n_{i++}$

V_3 : width $\sim$ cond freq: $V_3 | V_1, V_2 = n_{ijk} / n_{ij+}$

$\rightarrow$ Area $\sim$ cell frequency, n_{ijk}



Mosaic displays: basic ideas

Area proportional display for an n-way table

Tiles: recursive splits of a unit square, alternating H, V

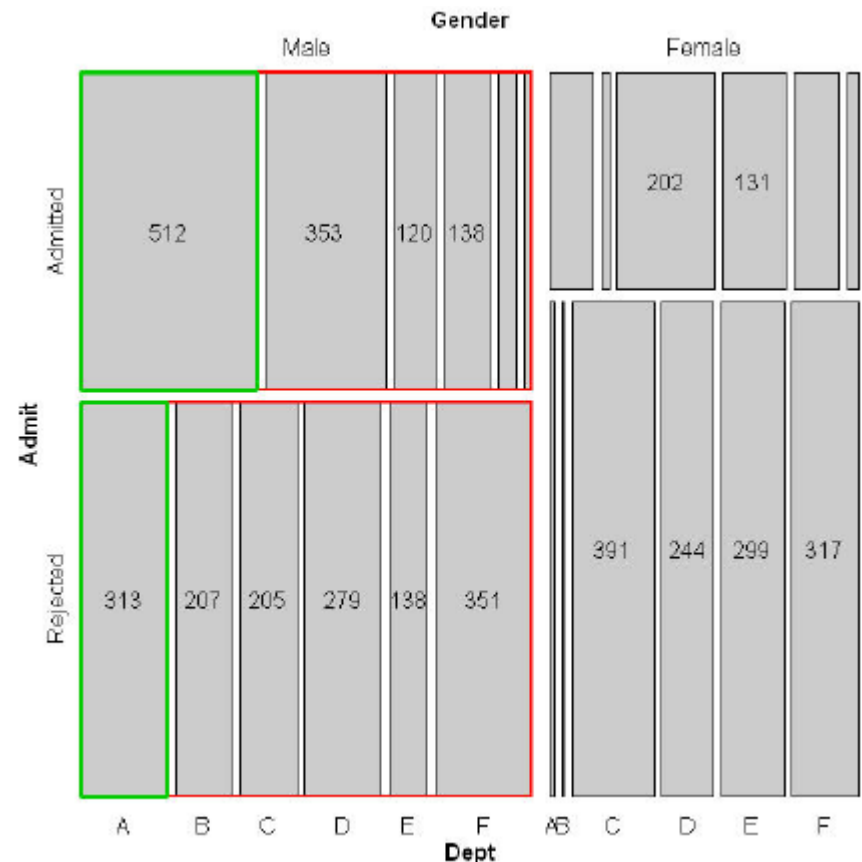
V_1 : width $\sim$ marginal frequencies, n_{i++}

V_2 : height $\sim$ cond freq: $V_2 | V_1 = n_{ij} / n_{i++}$

V_3 : width $\sim$ cond freq: $V_3 | V_1, V_2 = n_{ijk} / n_{ij+}$

$\rightarrow$ Area $\sim$ cell frequency, n_{ijk}

Gender x Admit x Dept frequencies



Mosaic displays: Independence

Expected frequencies if Admit $\perp$ Gender

Expected frequencies under independence are products of the row / col margins

$$\hat{m}_{ij} = \frac{n_{i+}n_{+j}}{n_{++}} = n_{++} \text{row \%col \%}$$

→ Row and col tiles align when variables are independent

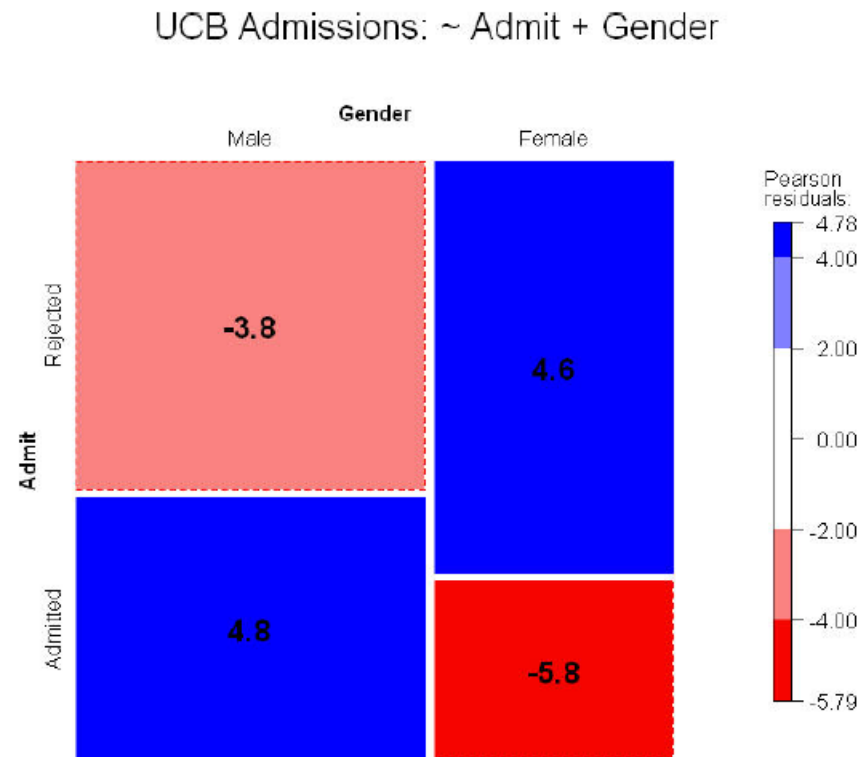


Mosaic displays: Residuals & shading

- Pearson residuals:

$$d_{ij} = \frac{n_{ij} - \hat{m}_{ij}}{\sqrt{\hat{m}_{ij}}}$$

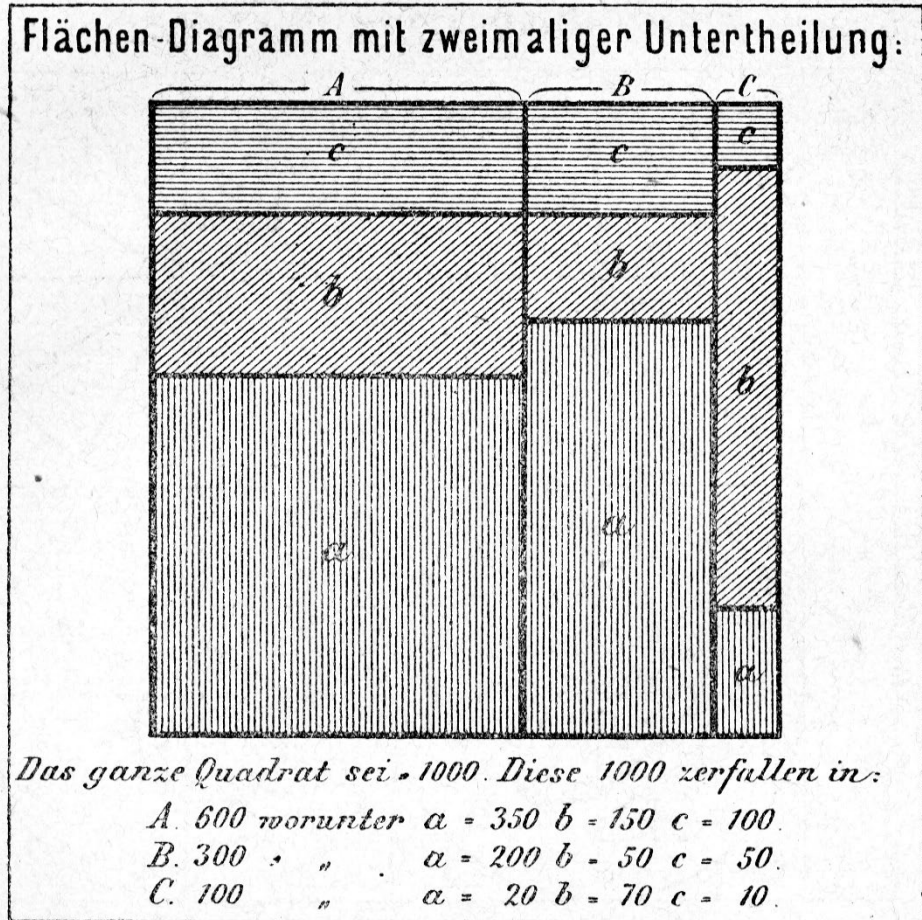
- Pearson $\chi^2 = \sum \sum d_{ij}^2 = \sum \sum \frac{(n_{ij} - \hat{m}_{ij})^2}{\hat{m}_{ij}}$
- Other residuals: deviance (LR), Freeman-Tukey (FT), adjusted (ADJ), ...
- Shading:
 - Sign: **— negative in red**; **+ positive in blue**
 - Magnitude: intensity of shading: $|d_{ij}| > 0, 2, 4, \dots$
- $\Rightarrow$ Independence: rows align, or cells are empty!



History corner: von Mayr



Georg von Mayr (1877) was the first to suggest an area-proportional display for two categorical variables



Figur 5.

Total count = 1000

Divided into (cows, pigs, sheep?)

A = 600, B = 300, C = 100

Each of these sub-divided by a 2nd variable (region: a, b, c)

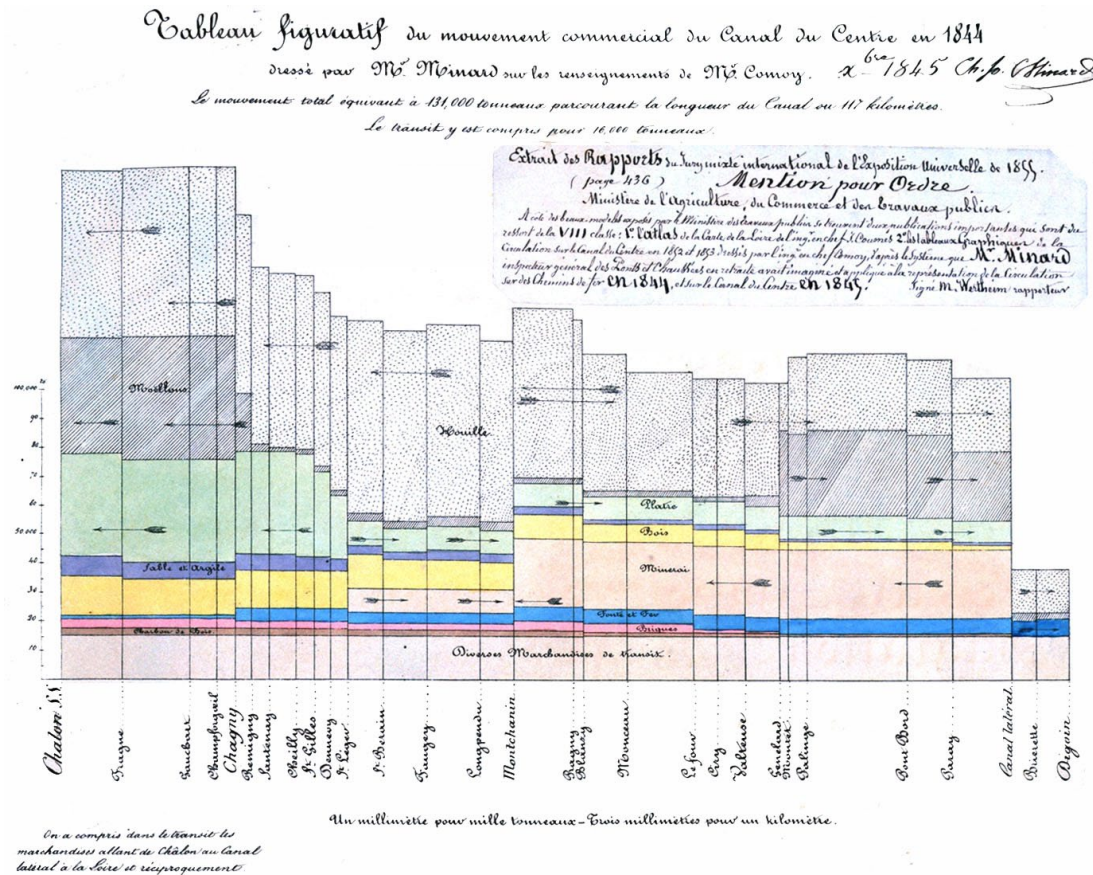
The name: "bottle diagram with double divisions" suggests further splits

See: Friendly (2002), "A Brief History of the Mosaic Display", *JCGS*, 11:1,89-107,
<http://dx.doi.org/10.1198/106186002317375631>

History corner: Minard

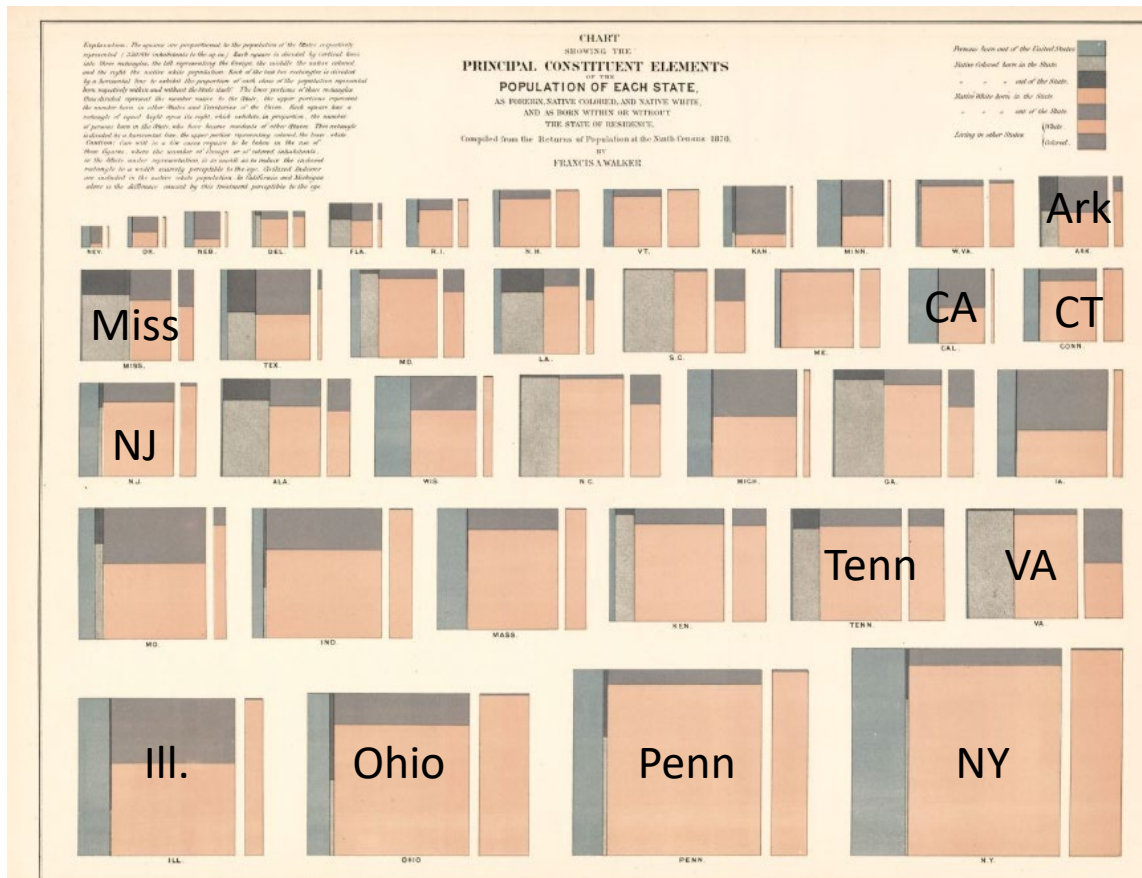
Charles Joseph Minard used an early form of an area-proportional plot to show the value of transport of goods along the Canal du Centre, from Chalon to Dijon.

- Width $\sim$ distance
 - Height $\sim$ amount of goods
- $\rightarrow \text{Value} = \text{distance} \times \text{amount}$



History corner: Francis Walker

In the 1870 *Statistical Album of the US Census*, Francis Walker wanted to show the populations of the states as comprised of {foreign, native colored, native white} and born {within, outside} their state of residence.



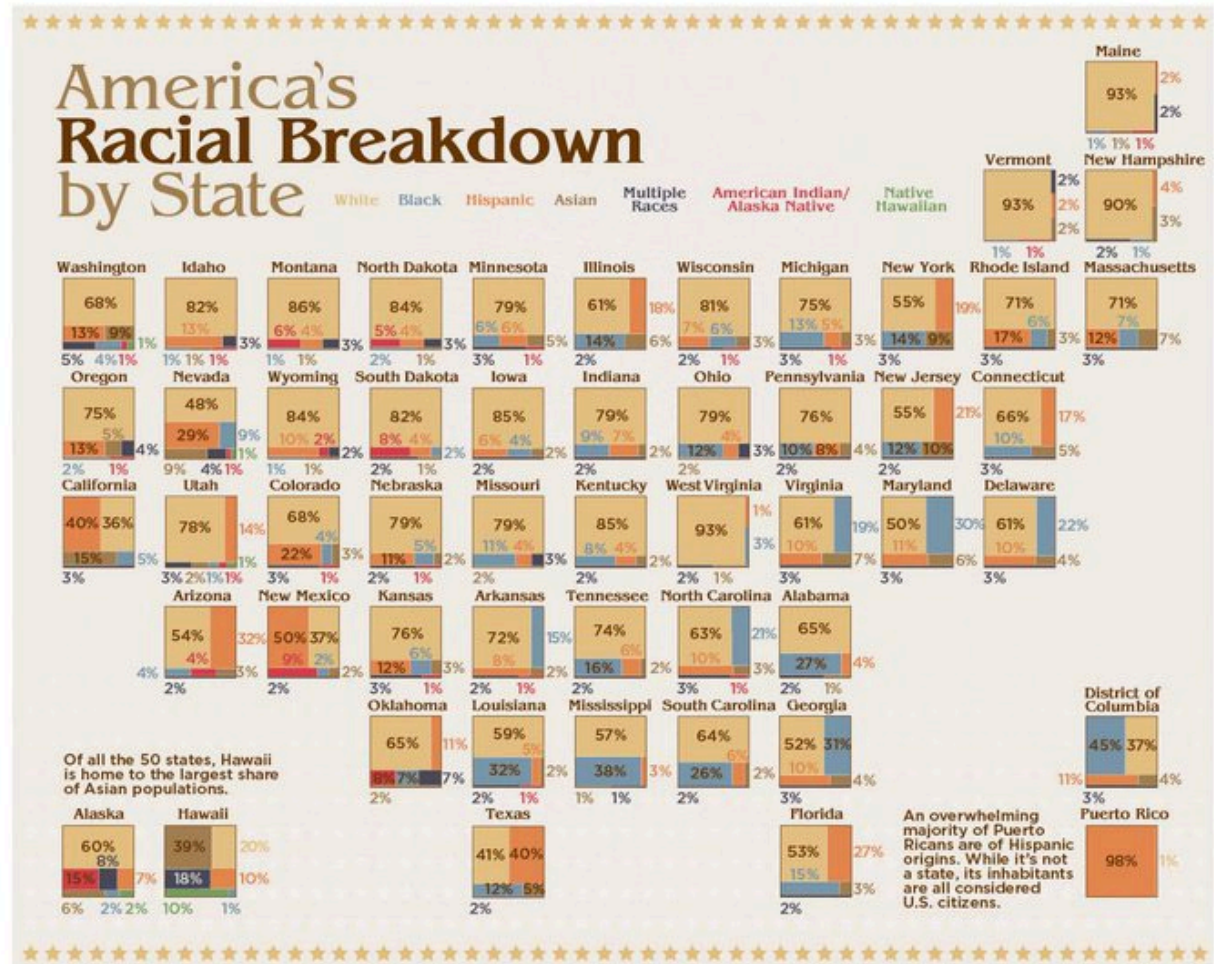
States are ordered by total pop, shown by area

For each state, a small mosaic, supplemented by a marginal one (w/in, outside)

Mosaic cartograms

US map provides a spatial framework for showing the distribution of categorical data

Each mosaic is positioned as in a schematic US map



Sources: Kaiser Family Foundation, U.S. Census Bureau

[f](#) [v](#) /visualcapitalist [t](#) [i](#) @visualcap [v](#) visualcapitalist.com

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Loglinear models: Perspectives



Loglinear models grew up and developed from three different ideas and **ways of thinking** about notions of independence in frequency data

- **Loglinear approach:** analog of ANOVA; associations are $\sim$ interactions
- **glm() approach:** analog of general regression model, for $\log(\text{Freq})$, with Poisson dist^n of errors
- **Logit models:** Loglinear, simplified for a binary response

Loglinear approach

First developed as analog of classical ANOVA models, where multiplicative relations are re-expressed in additive form as models for $\log(\text{Freq})$

$$\log m_{ij} = \mu + \lambda_i^A + \lambda_j^B \equiv [A][B] \equiv \sim A + B$$

- This expresses the **independence** model for a 2-way table as no A*B association
- Short-hand notations: $[A][B] = A \perp B = \sim A + B$
- Fit by simple **iterative proportional scaling**: **MASS::loglm()**
 - Parameters aren't estimated; only fitted frequencies.

```
loglm(Freq ~ A + B + C)      # [A] [B] [C]
loglm(Freq ~ A * B + C )    # [A B] [C]
loglm(Freq ~ A * B * C)     # [A B C]
```

glm() approach

Extension of classical linear models recognized loglinear models as a model for $\log(\text{Freq})$, with Poisson dist^n for cell counts

$$\log \mathbf{m} = \mathbf{X} \boldsymbol{\beta}$$

- Looks like std ANOVA/regression model, but for $\log(\text{Freq})$
- This allows **quantitative** predictors and special ways to treat **ordinal** factors
- Fit by **maximum likelihood** using `glm(..., family=poisson)`
 - Can estimate parameters; do structured tests
- Standard diagnostic methods available

```
glm( Freq ~ A + B + C, family = poisson)      # [A] [B] [C]
glm( Freq ~ A * B + C, family = poisson)      # [A B] [C]
glm( Freq ~ A * (B+C), family = poisson)      # [A B] [A C]
```

Logit models

When one variable is a **binary** response, a logit model is a simpler way to specify a loglinear model

$$\log(m_{1jk}/m_{2jk}) = \alpha + \beta_j^B + \beta_k^C \equiv [AB][AC][BC]$$


- $\log(m_{1jk}/m_{2jk})$ is the **log odds** of response A1 vs A2
- The model only includes terms for the effect of A on B & C
- Equivalent loglinear model: $[AB][AC]$ **$[BC]$**
- The logit models **assumes** the $[BC]$ association;

$$[AB] \rightarrow \beta_j^B \quad [AC] \rightarrow \beta_k^C$$

- Fit using **family=binomial**

```
glm(outcome=="survived" ~ B + C, family = binomial)
```


Two-way tables: loglinear approach

For two discrete variables, A and B , suppose a multinomial sample of total size n over the IJ cells of a two-way $I \times J$ contingency table, with cell frequencies n_{ij} , and cell probabilities $\pi_{ij} = n_{ij}/n$.

- The table variables are **statistically independent** when the cell (joint) probability equals the product of the marginal probabilities, $\Pr(A = i \ \& \ B = j) = \Pr(A = i) \times \Pr(B = j)$, or,

$$\pi_{ij} = \pi_{i+} \pi_{+j} \ .$$

- An equivalent model in terms of expected frequencies, $m_{ij} = n\pi_{ij}$ is

$$m_{ij} = (1/n) \ m_{i+} \ m_{+j} \ .$$

- This multiplicative model can be expressed in additive form as a model for $\log m_{ij}$,

$$\log m_{ij} = -\log n + \log m_{i+} + \log m_{+j} \ . \quad (1)$$

Two-way tables: loglinear approach

Independence model

By analogy with ANOVA models, the independence model (1) can be expressed as

$$\log m_{ij} = \mu + \lambda_i^A + \lambda_j^B, \quad (2)$$

- μ is the grand mean of $\log m_{ij}$
- the parameters λ_i^A and λ_j^B express the marginal frequencies of variables A and B — “main effects”
- typically defined so that $\sum_i \lambda_i^A = \sum_j \lambda_j^B = 0$ as in ANOVA

Two-way tables: loglinear approach

Saturated model

Dependence between the table variables is expressed by adding association parameters, λ_{ij}^{AB} , giving the *saturated model*,

$$\log m_{ij} = \mu + \lambda_i^A + \lambda_j^B + \lambda_{ij}^{AB} \equiv [AB] \equiv \sim A * B . \quad (3)$$

- The saturated model fits the table perfectly ($\hat{m}_{ij} = n_{ij}$): there are as many parameters as cell frequencies. Residual df = 0.
- A global test for association tests $H_0 : \lambda_{ij}^{AB} = 0$.
- If reject H_0 , which $\lambda_{ij}^{AB} \neq 0$?
- For *ordinal* variables, the λ_{ij}^{AB} may be structured more simply, giving tests for ordinal association.

Example: Independence

Generate a table of Education by Party preference, strictly independent

```
> educ <- c(50, 100, 50) # marginal frequencies
> names(educ) <- c("Low", "Med", "High")
> party <- c(20, 50, 30) # marginal frequencies
> names(party) <- c("NDP", "Liberal", "Cons")
> table <- outer(educ, party) / sum(party) # cell = row * col / n
> names(dimnames(table)) <- c("Education", "Party")
> table
```

	Party		
Education	NDP	Liberal	Cons
Low	10	25	15
Med	20	50	30
High	10	25	15

Perfect fit:

```
> MASS::loglm(~ Education + Party, table)
Call:
MASS::loglm(formula = ~Education + Party, data = table)

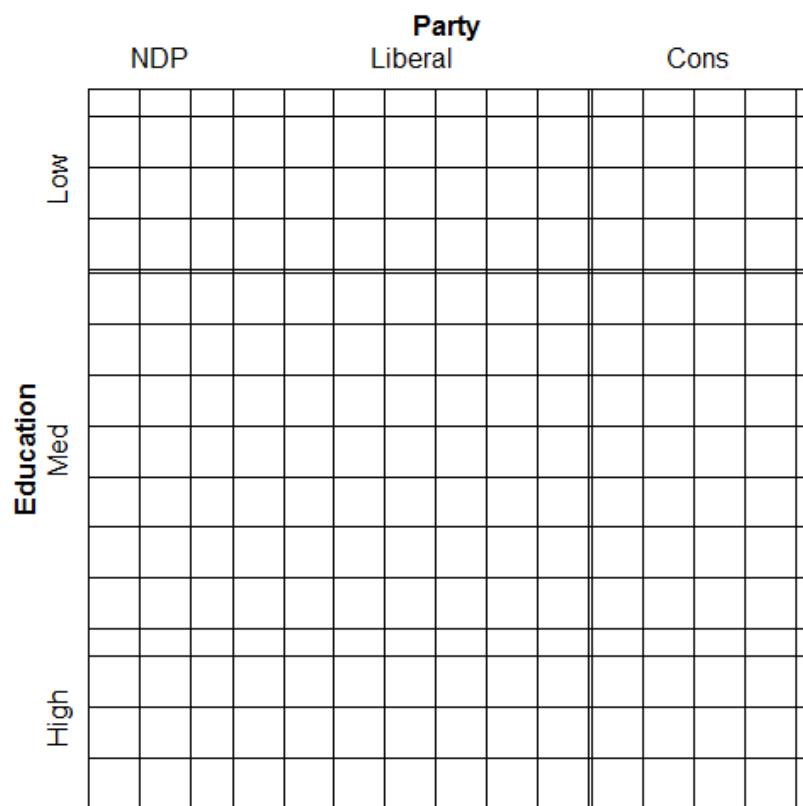
Statistics:
```

	X^2	df	P(> X^2)
Likelihood Ratio	0	4	1
Pearson	0	4	1

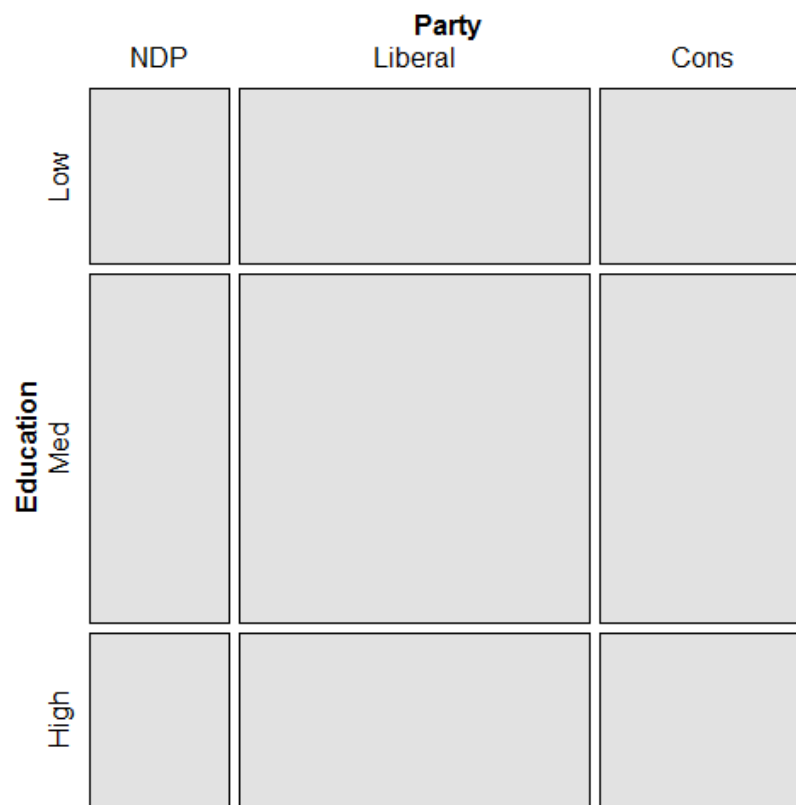


Both sieve diagrams and mosaic plots show what independence “looks like”

```
> sieve(table, shade=TRUE)
```



```
> mosaic(table, shade=TRUE)
```



Two-way tables: glm approach

In the GLM approach, the vector of cell frequencies, $\mathbf{n} = \{n_{ij}\}$ is specified to have a **Poisson** distribution with means $\mathbf{m} = \{m_{ij}\}$ given by

$$\log \mathbf{m} = \mathbf{X}\beta$$

- $\mathbf{X}$ is a known design (model) matrix, expressing the table factors
 - β is a column vector containing the unknown λ parameters.
-
- This is the same as the familiar matrix formulation of ANOVA/regression, except that
 - The response, $\log \mathbf{m}$ makes multiplicative relations additive
 - The distribution is taken as Poisson rather than Gaussian (normal)

Example: 2 x 2 table

For a 2×2 table, the saturated model (3) with the usual zero-sum constraints can be represented as

$$\log \begin{pmatrix} m_{11} \\ m_{12} \\ m_{21} \\ m_{22} \end{pmatrix} = \begin{matrix} X \\ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \end{matrix} \begin{matrix} \beta \\ \begin{pmatrix} \mu \\ \lambda_1^A \\ \lambda_1^B \\ \lambda_{11}^{AB} \end{pmatrix} \end{matrix} \quad \begin{matrix} \text{total } n \\ \text{margin A} \\ \text{margin B} \\ \text{association} \end{matrix}$$

- only the linearly independent parameters are represented. $\lambda_2^A = -\lambda_1^A$, because $\lambda_1^A + \lambda_2^A = 0$, and so forth.
- association** is represented by the parameter λ_{11}^{AB}
- can show that $\lambda_{11}^{AB} = \frac{1}{4} \log(\theta)$ (log odds ratio)

- Advantages of the GLM formulation: easier to express models with ordinal or quantitative variables, special terms, etc. Can also allow for **over-dispersion**.

i.e., $\text{Var}(m_{ij}) > m_{ij}$ then can use NegBin

Assessing goodness of fit

Goodness of fit of a specified model may be tested by the likelihood ratio G^2 ,

$$G^2 = 2 \sum_i n_i \log \left(\frac{n_i}{\hat{m}_i} \right) , \quad (4)$$

or the Pearson X^2 ,

$$X^2 = \sum_i \frac{(n_i - \hat{m}_i)^2}{\hat{m}_i} , \quad (5)$$

with degrees of freedom $df = \# \text{ cells} - \# \text{ estimated parameters}$.

- E.g., for the model of independence, $[A][B]$, $df = IJ - [(I - 1) + (J - 1)] = (I - 1)(J - 1)$
- The terms summed in (4) and (5) are the squared *cell residuals*
- Other measures of balance goodness of fit against parsimony, e.g., *Akaike's Information Criterion* (smaller is better)

$$AIC = G^2 - 2df \text{ or } AIC = G^2 + 2 \# \text{ parameters}$$

Degrees of freedom

Q: Why do degrees of freedom for an $r \times c$ table = $(r-1) * (c-1)$?

A: Because the row & column totals are used to calculate expected frequencies

→ Only $(r-1)$ independent parameters for each column

→ Only $(c-1)$ independent parameters for each row

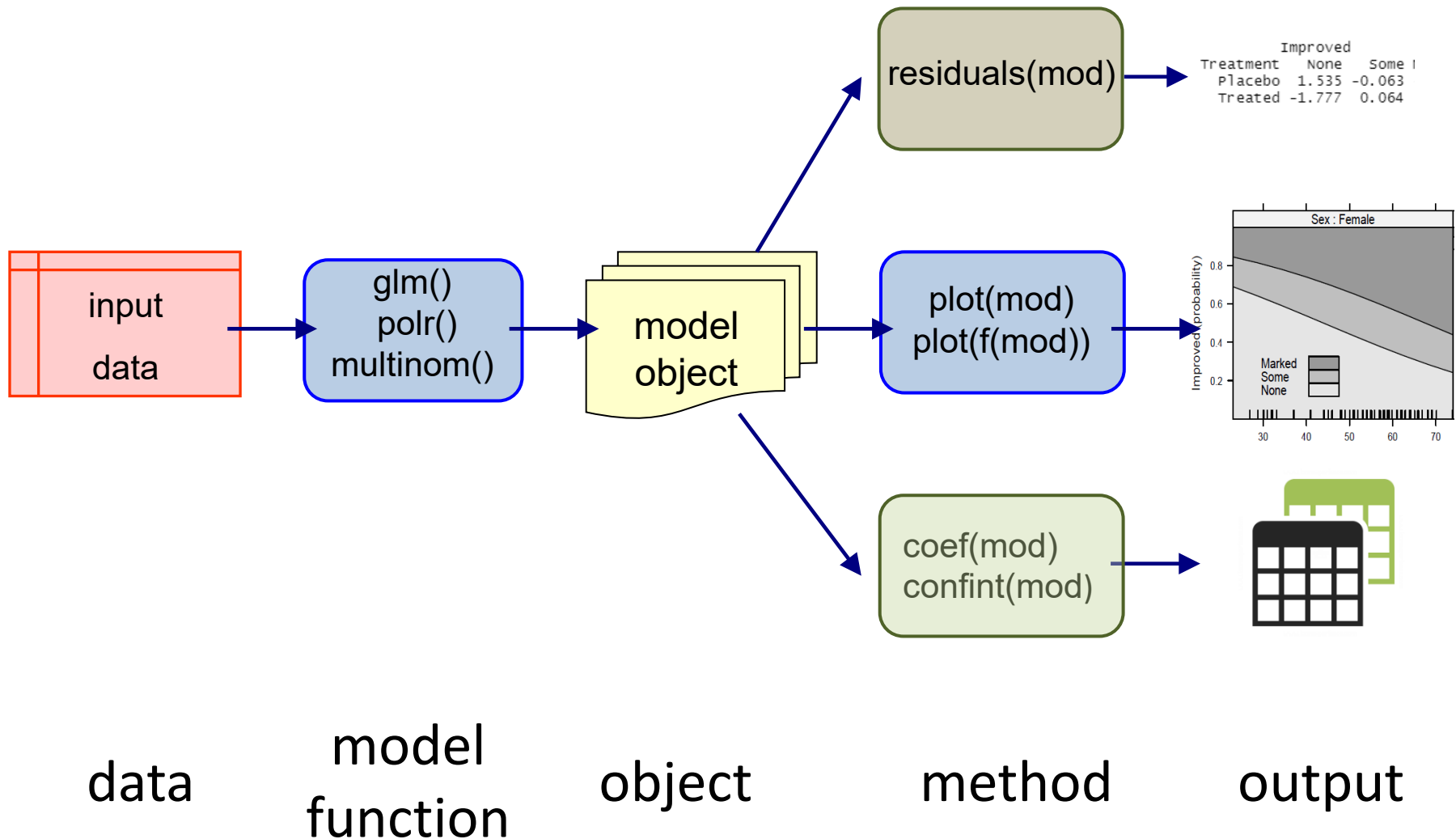
```
> addmargins(HE) |> knitr::kable()
```

	Black	Brown	Red	Blond	Sum
:	:	:	:	:	:
Brown	68	119	26	7	220
Blue	20	84	17	94	215
Hazel	15	54	14	10	93
Green	5	29	14	16	64
:	:	:	:	:	:
Sum	108	286	71	127	592

R functions for loglinear models

- **vcd::assocstats()** – only χ^2 tests for two-way tables; not a model (no parameters; no residuals)
- **MASS::loglm()** – general loglinear models for n -way tables
`loglm(formula, data, subset, na.action, ...)`
- **glm()** – all generalized linear models; loglinear with **family = poisson**
`glm(formula, data, weights, subset, ...)`
- Model formulas have the form:
 - **table** form: $\sim A + B + \dots$ (independence);
 - $\sim A * B + C$ (allow $A*B$ association)
 - **frequency** data frame: $\text{Freq} \sim A * B + C$

Model-based methods: Fitting & graphing



R functions & methods

- `loglm()` and `glm()` return an R object with named components and with a `class()` – here “loglm”

```
> arth.mod <- loglm(~Treatment+Improved, data=arth.tab, fitted=TRUE)

> names(arth.mod)
[1] "lrt"          "pearson"      "df"           "margin"       "fitted"       "param"
[7] "call"         "formula"      "frequencies"  "deviance"     "nobs"         "terms"

class(arth.mod)
[1] "loglm"
```

- Objects have **methods**: `print()`, `summary()`, `coef()`, `residuals()`, `plot()` and other methods
 - Methods are specific to the class of the object
 - `print(arth.mod) → print.loglm(arth.mod)`
 - `residuals(arth.mod) → residuals.loglm(arth.mod)`

Example: Arthritis treatment

Data on effects of treatment for rheumatoid arthritis (in [case form](#))

```
> data(Arthritis, package="vcd")
> str(Arthritis)
'data.frame':      84 obs. of  5 variables:
 $ ID      : int  57 46 77 17 36 23 75 39 33 55 ...
 $ Treatment: Factor w/ 2 levels "Placebo","Treated": 2 2 2 2 2 2 2 2 2 2 ...
 $ Sex      : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 2 2 ...
 $ Age      : int  27 29 30 32 46 58 59 59 63 63 ...
 $ Improved : Ord.factor w/ 3 levels "None"<"Some"<...: 2 1 1 3 3 3 1 3 1 1 ...
```

For now, ignore Age; consider the 2 x 3 table of Treatment x Improved

```
> arth.tab <- with(Arthritis, table(Treatment, Improved))
> arth.tab
```

	Improved		
Treatment	None	Some	Marked
Placebo	29	7	7
Treated	13	7	21

arth.tab is a "table" object
print.table() gives this output

Arthritis treatment

Fit the independence model, $\sim$ Treatment + Improved

```
> (arth.mod <- loglm(~Treatment + Improved, data = arth.tab, fitted=TRUE))  
Call:  
loglm(formula = ~Treatment + Improved, data = arth.tab, fitted = TRUE)
```

Statistics:

	X^2	df	P(> X^2)
Likelihood Ratio	13.53	2	0.001154
Pearson	13.06	2	0.001463

The method `print.loglm()` gives this output

Some methods: `residuals()`, `deviance()`, `coef()`, ...

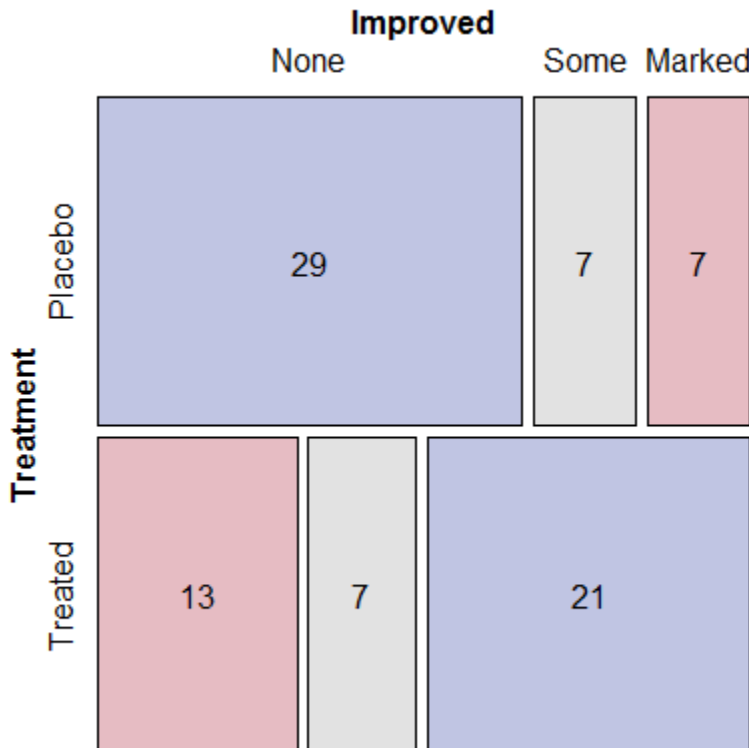
```
> round(residuals(arth.mod), 3)  
      Improved  
Treatment  None   Some Marked  
  Placebo  1.535 -0.063 -2.152  
  Treated -1.777  0.064  1.837  
  
# Likelihood ratio chisquare  
> deviance(arth.mod)  
[1] 13.53
```

```
> coef(arth.mod)  
$`(Intercept)`  
[1] 2.543  
  
$Treatment  
  Placebo  Treated  
  0.0238  -0.0238  
  
$Improved  
      None      Some      Marked  
  0.5014  -0.5972   0.0959
```

Arthritis treatment: Plots

Visualization: `mosaic()` or `plot()` the model or table

```
> mosaic(arth.mod, shade=TRUE, gp_args=list(interpolate=1:4),  
         labeling = labeling_values)
```



Pearson
residuals:
2.0
1.0
0.0
-1.0
-1.9
p-value =
0.0015

Splits by the response,
Treatment first

Custom scheme for
shading levels; normally
`c(2, 4)` for `|residual|`

Cells can be labeled by
freq, residual, ...

Arthritis treatment: glm()

glm() for loglinear models easiest w/ the data as a data.frame in frequency form, `family=poisson`

```
> arth.df <- as.data.frame(xtabs(~ Treatment + Improved,
                                data=Arthritis))
```

```
> arth.df
```

	Treatment	Improved	Freq
1	Placebo	None	29
2	Treated	None	13
3	Placebo	Some	7
4	Treated	Some	7
5	Placebo	Marked	7
6	Treated	Marked	21

```
> arth.glm <- glm(Freq ~ Treatment + Improved, data = arth.df,
                  family = poisson)
```

More on `glm()` models later

Example: Hair color & Eye color

```
> haireye <- margin.table(HairEyeColor, 1:2)
> (HE.mod <- loglm(~ Hair + Eye, data=haireye))
Call:
loglm(formula = ~Hair + Eye, data = haireye)
```

Statistics:

	X^2	df	P(> X^2)
Likelihood Ratio	146.4	9	0
Pearson	138.3	9	0

```
> round(residuals(HE.mod), 2)
```

Re-fitting to get frequencies and fitted values

	Eye			
Hair	Brown	Blue	Hazel	Green
Black	4.00	-3.39	-0.49	-2.21
Brown	1.21	-2.02	1.31	-0.35
Red	-0.08	-1.85	0.82	2.04
Blond	-7.33	6.17	-2.47	0.60

New in vcdExtra: color_table()



Idea: Semi-graphic “heatmap” display of a frequency table or residuals, shading background ~ residuals (remaining association, not fit by a model)

```
color_table(haireye)
```

	Eye				Total
	Brown	Blue	Hazel	Green	
Black	68	20	15	5	108
Brown	119	84	54	29	286
Red	26	17	14	14	71
Blond	7	94	10	16	127
Total	220	215	93	64	592

```
color_table(haireye,  
            values = "residuals")
```

	Eye			
	Brown	Blue	Hazel	Green
Black	4.40	-3.07	-0.48	-1.95
Brown	1.23	-1.95	1.35	-0.35
Red	-0.07	-1.73	0.85	2.28
Blond	-5.85	7.05	-2.23	0.61

Mosaic displays: Seeing patterns



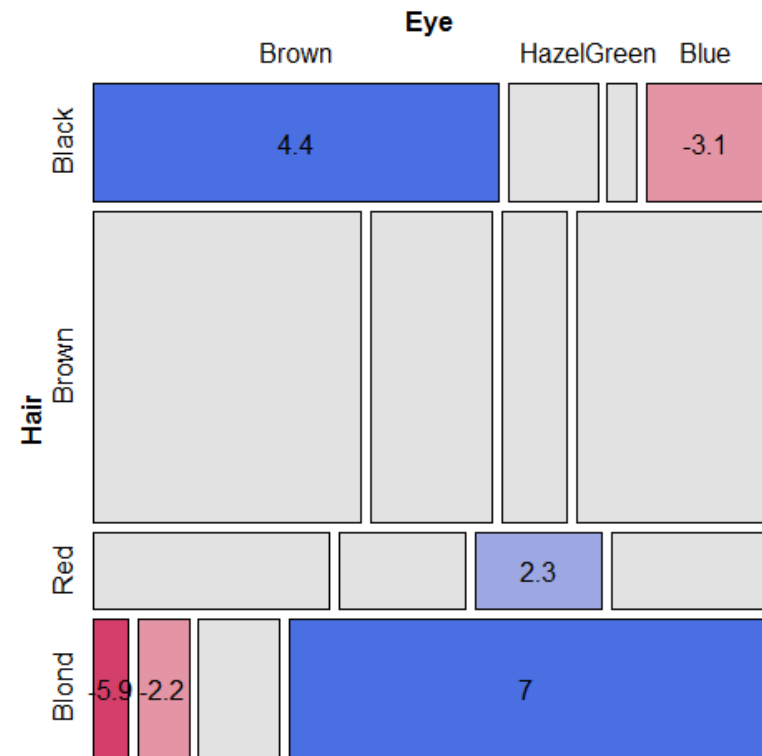
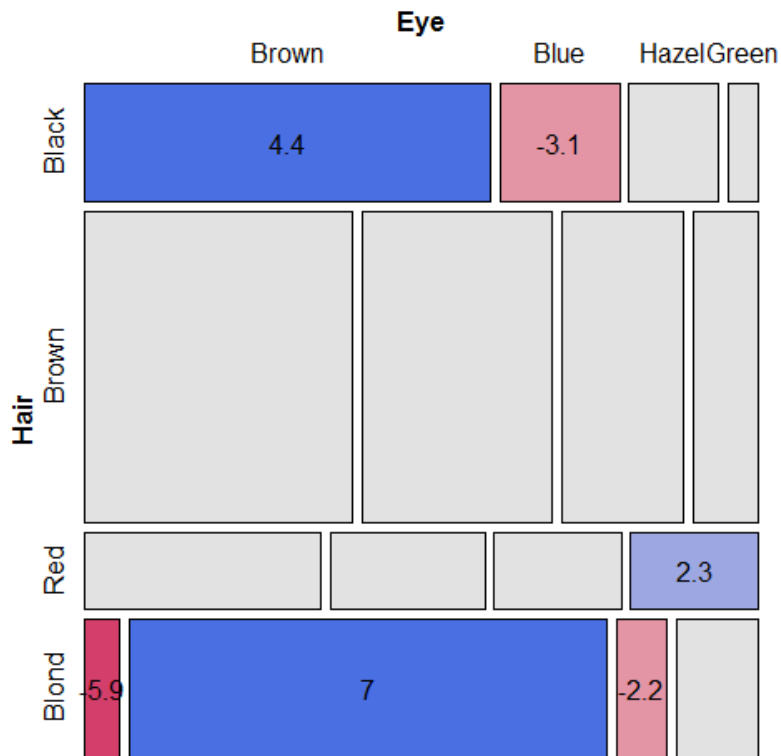
- In two-way models, residuals contain the info on lack of independence
 - Equivalently: help to understand the **pattern** of association
 - **Effect ordering**: permuting the rows / cols often makes the pattern more apparent
- Correspondence analysis: → reorder by scores on Dim 1
 - `seriation::permute(order="CA")` does this for two-way tables

```
> haireye
      Eye
Hair   Brown Blue Hazel Green
Black    68   20   15    5
Brown   119   84   54   29
Red      26   17   14   14
Blond     7   94   10   16
```

```
> library(seriation)
> permute(haireye, "CA")
      Eye
Hair   Brown Hazel Green Blue
Black    68   15    5   20
Brown   119   54   29   84
Red      26   14   14   17
Blond     7   10   16   94
```

Mosaic displays: Seeing patterns

```
mosaic(haireye, shade=TRUE, labeling=labeling_residuals)
mosaic(permute(haireye, "CA"), shade=TRUE, labeling=labeling_residuals)
```





Bee abundance data



A study by Taylor Kerekes examined the abundance of bee species in Ontario over three periods of time.

Q: Does relative abundance of species differ over years?

A: Do a chi-square test

```
chisq.test(bees[, -1])  
Pearson's Chi-squared test
```

```
data: bees[, -1]  
X-squared = 1981, df = 26, p-value <2e-16
```

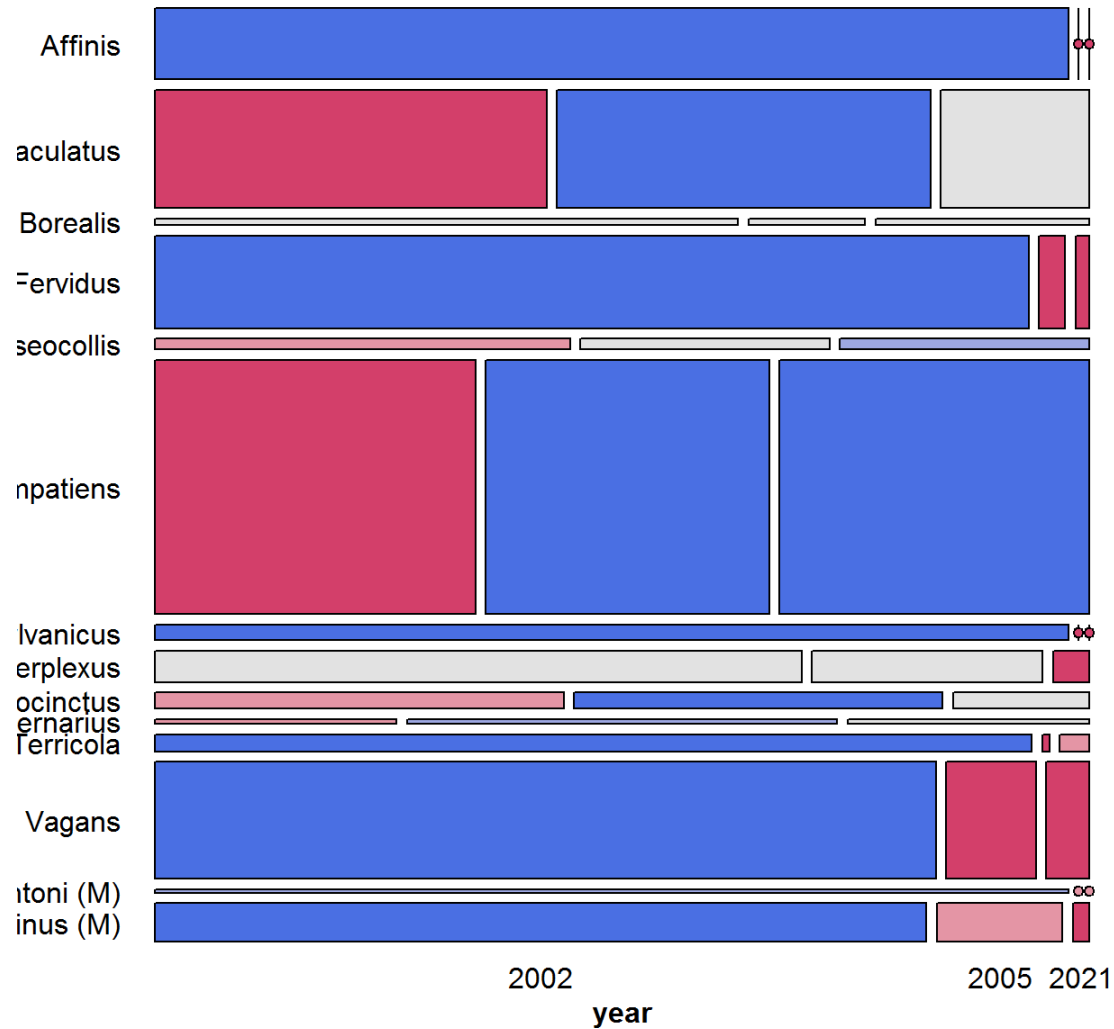


# A tibble: 14 x 4				
	species	`2002`	`2005`	`2021`
	<chr>	<dbl>	<dbl>	<dbl>
1	Affinis	508	0	0
2	Bimaculatus	362	345	137
3	Borealis	30	6	11
4	Fervidus	634	19	10
5	Griseocollis	35	21	21
6	Impatiens	638	564	616
7	Pensylvanicus	112	0	0
8	Perplexus	160	57	9
9	Rufocinctus	51	46	17
10	Ternarius	9	16	9
11	Terricola	119	1	4
12	Vagans	713	82	39
13	Ashtoni (M)	27	0	0
14	Citrinus (M)	234	38	5

How to understand the pattern of association?

```
mosaic(bees.mat, shade=TRUE, ...)
```

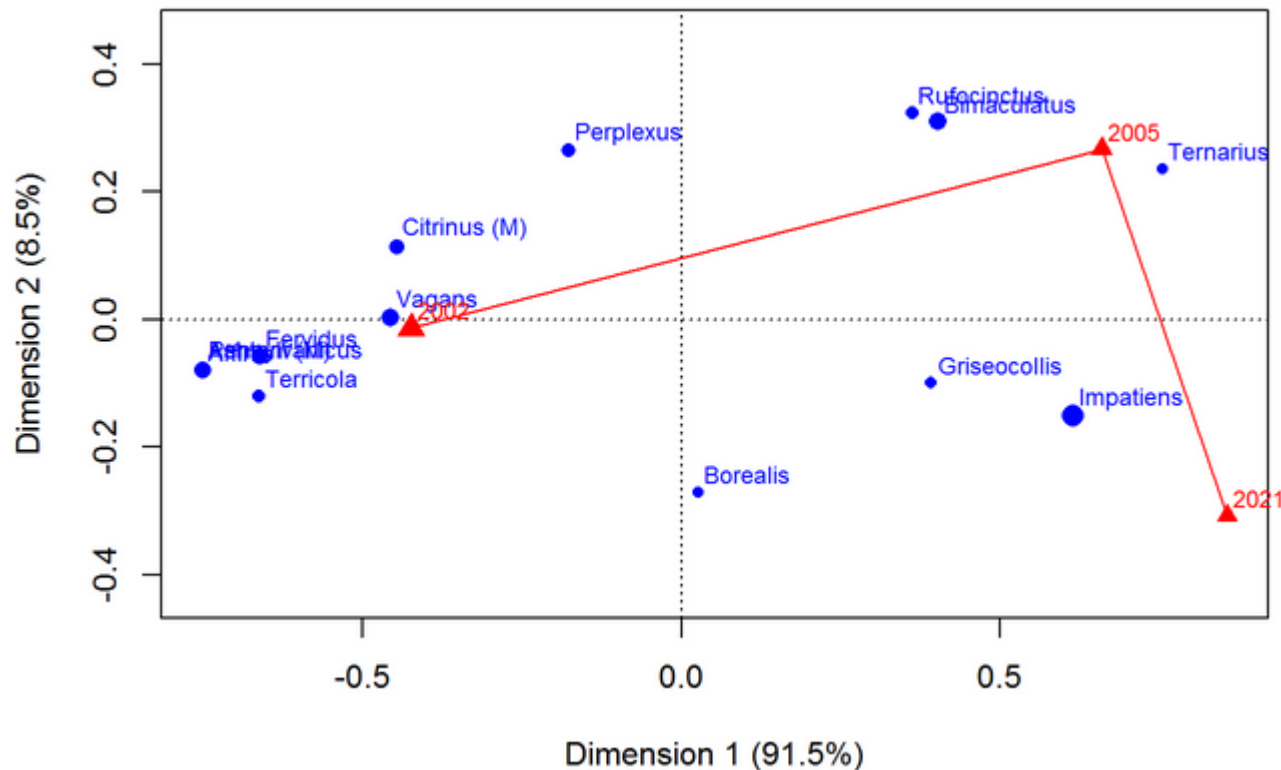
Bees Abundance Data



Alphabetic order of
species:
No clear pattern

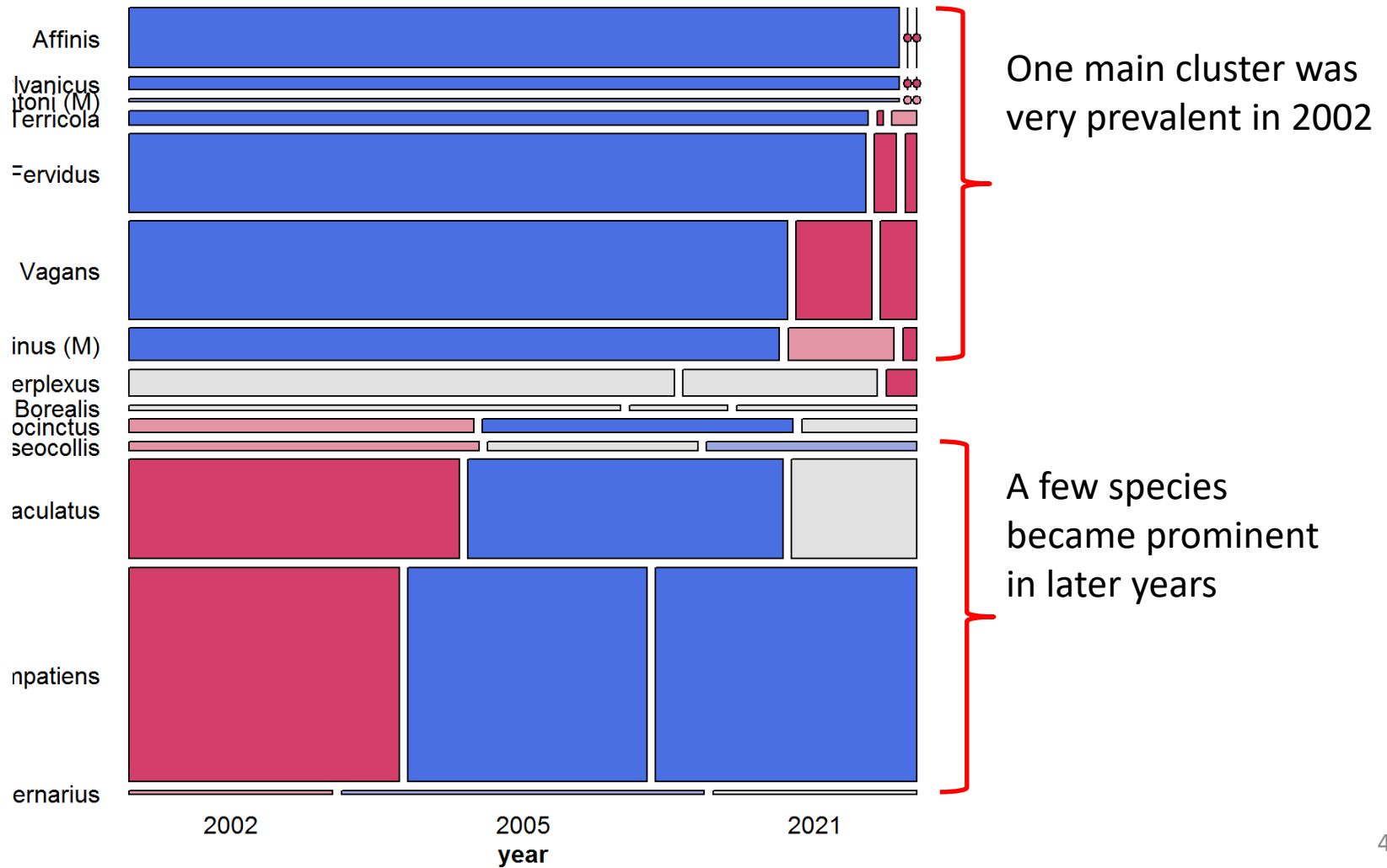
Correspondence analysis finds scores for the row & col categories to account for maximum χ^2

```
bees.ca <- ca(bees.mat)
plot(bees.ca,
     lines=c(FALSE, TRUE),      # join years with lines
     mass = c(TRUE, TRUE))     # symbol size ~ marginal frequency
```




```
mosaic(permute(bees.mat, "CA"), shade=TRUE, ...)
```

Bees Abundance Data



Three-way tables

Saturated model

For a 3-way table, of size $I \times J \times K$ for variables A, B, C , the **saturated** loglinear model includes associations between all pairs of variables, as well as a 3-way association term, λ_{ijk}^{ABC}

$$\log m_{ijk} = \mu + \lambda_i^A + \lambda_j^B + \lambda_k^C + \lambda_{ij}^{AB} + \lambda_{ik}^{AC} + \lambda_{jk}^{BC} + \lambda_{ijk}^{ABC}. \quad (6)$$

- One-way terms ($\lambda_i^A, \lambda_j^B, \lambda_k^C$): differences in the *marginal frequencies* of the table variables.
- Two-way terms ($\lambda_{ij}^{AB}, \lambda_{ik}^{AC}, \lambda_{jk}^{BC}$) pertain to the *partial association* for each pair of variables, *controlling* for the remaining variable.
- The three-way term, λ_{ijk}^{ABC} allows the partial association between any pair of variables to vary over the categories of the third variable.
- Fits perfectly, but doesn't *explain* anything, so we hope for a simpler model!

Reduced models

- Goal: fit the **smallest** model sufficient to explain/describe the observed frequencies
 - Similar to Anova models, e.g., $\sim(A + B + C)^3$ with all interactions
 - Do we need all those interaction terms?
- **Hierarchical** models
 - Most loglinear models are taken to be hierarchical:
 - A high-order term, like $\lambda_{ijk}^{ABC} \rightarrow$ all **lower order terms** included
 - Why: principle of **marginality**– hard to interpret w/o low order relatives
- Thus, a shorthand notation for a loglinear model lists only the **high-order** terms. For example:
 - $[ABC] \rightarrow A + B + C + AB + AC + BC + ABC$
 - $[AB][AC] \rightarrow A + B + C + AB + AC$

Reduced models

- For a three-way table there is a range of models between mutual independence, $[A][B][C]$, and the saturated model, $[ABC]$
- Each model has an independence interpretation:

$$[A][B] \equiv A \perp B \equiv A \text{ independent of } B$$

- Special names for various submodels

Table: Log-linear Models for Three-Way Tables

Model	Model symbol	Interpretation
Mutual independence	$[A][B][C]$	$A \perp B \perp C$
Joint independence	$[AB][C]$	$(A \ B) \perp C$
Conditional independence	$[AC][BC]$	$(A \perp B) \mid C$
All two-way associations	$[AB][AC][BC]$	homogeneous assoc.
Saturated model	$[ABC]$	ABC interaction

Model types

- **Joint independence:** $(AB) \perp C$, allows $A*B$ association, but asserts no $A*C$ and $B*C$ associations

$$[AB][C] \equiv \log m_{ijk} = \mu + \lambda_i^A + \lambda_j^B + \lambda_k^C + \lambda_{ij}^{AB}$$

- **Conditional independence:** $A \perp B$, controlling for C

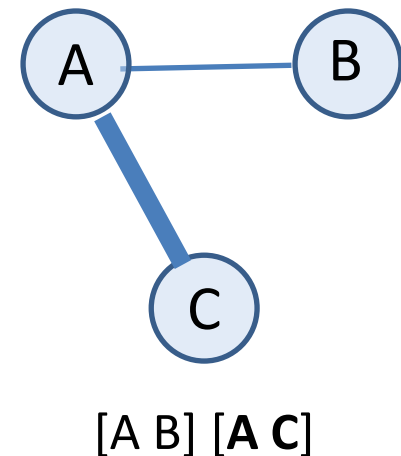
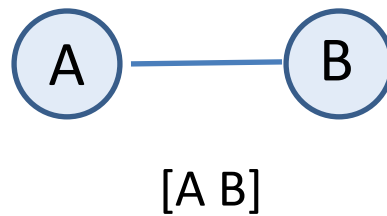
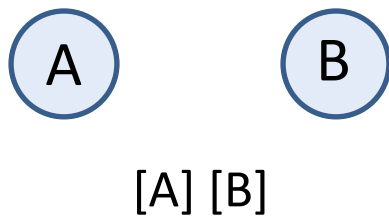
$$[AC][BC] \equiv \log m_{ijk} = \mu + \lambda_i^A + \lambda_j^B + \lambda_k^C + \lambda_{ik}^{AC} + \lambda_{jk}^{BC}$$

- **Homogeneous association:** All two-way, but each two-way is the *same* over the other factor

$$[AB][AC][BC] \equiv \log m_{ijk} = \mu + \lambda_i^A + \lambda_j^B + \lambda_k^C + \lambda_{ij}^{AB} + \lambda_{ik}^{AC} + \lambda_{jk}^{BC}$$

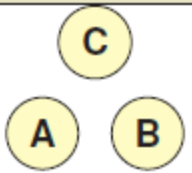
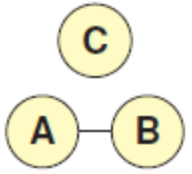
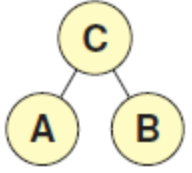
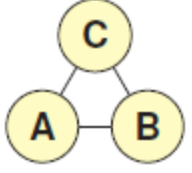
Association graphs

- An association graph represents variables in an n -way frequency table by an **undirected graph**
 - Nodes are the variables
 - Edges are first-order (2-way) **associations**
 - → two variables are **independent** if not joined by an edge



Under-solved problem: Use an association graph to represent **strength** of conditional associations or as a widget for model fitting

Model types & association graphs

Hypothesis	Fitted margins	Model symbol	Independence interpretation	Association graph
H_1	$n_{i++}, n_{+j+}, n_{++k}$	$[A][B][C]$	$A \perp B \perp C$	
H_2	n_{ij+}, n_{++k}	$[AB][C]$	$(A, B) \perp C$	
H_3	n_{i+k}, n_{+jk}	$[AC][BC]$	$A \perp B \mid C$	
H_4	$n_{ij+}, n_{i+k}, n_{+jk}$	$[AB][AC][BC]$	NA	

Model types: loglm()

Each of these have simple translations into the model formulae for loglm()

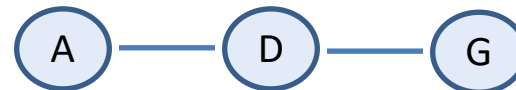
<code>loglm(~ A + B + C)</code>	# mutual independence	[A] [B] [C]
<code>loglm(~ A * B + C)</code>	# joint independence	[AB] [C]
<code>loglm(~ A*C + B*C)</code>	# conditional independence	[AC] [BC]
<code>loglm(~ (A + B + C)^2)</code>	# homogeneous, all 2-way	[AB] [AC] [BC]
<code>loglm(~ A * B * C)</code>	# saturated model	[ABC]

e.g., Berkeley data

`loglm(~ (Admit + Gender) * Dept)`

`loglm(~ Admit*Dept + Gender * Dept)`

Association graph



$\rightarrow A \perp G \mid D$

(just put your finger over D)

Collapsibility: Marginal & conditional associations

- Q: When can we legitimately collapse a table, $\{ABC\}$, over some variable (C)?
- A: When the **marginal** association of AB is the same as the **conditional** association, $AB \mid C$
- Recall the Berkeley data
 - Margin of [Admit, Gender] ignoring Dept showed strong association
 - The partial assoc. within Dept were mostly NS
 - Conditional association [Admit, Gender $\mid$ Dept] $\neq$ marginal [A, G]
- Three-way tables: The AB marginal and $AB \mid C$ conditional associations are the same, if either:
 - A & C are conditionally independent, $A \perp C \mid B = [AB][CB]$
 - B & C are conditionally independent, $B \perp C \mid A = [AB][AC]$
 - $\rightarrow$ no three-way association

Higher-way tables

DaytonSurvey data: A 2^5 table

2,276 HS seniors asked if they had ever used cigarettes, alcohol, or marijuana

		cigarette		alcohol		marijuana			
		Yes	No	Yes	No	Yes	No	Yes	No
sex	race	Yes	No	Yes	No	Yes	No	Yes	No
female	white	405	268	1	17	13	218	1	117
	other	23	23	0	1	2	19	0	12
male	white	453	228	1	17	28	201	1	133
	other	30	19	1	8	1	18	0	17

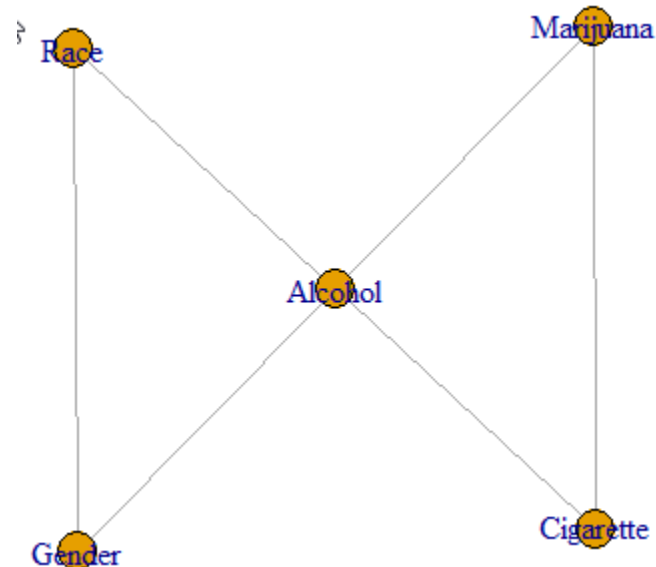
Suppose we wish to fit the model:

[A M] [A C] [M C] [A R] [A G] [R G]

The association graph implies:

$\{\text{race, gender}\} \perp \{\text{marijuana, cigarette}\} \mid \text{alcohol}$

If it fits, we can collapse the table over {race, gender} to study associations among A, M & C.



Response vs. Association models

- In **association models**, the interest is just on *which* variables are associated, and *how*
 - Hair-eye data: [Hair Eye]? [Hair Sex]? [Eye Sex]
 - $\implies$ fit the homogeneous association model (or the saturated model)
 - Test the individual terms, delete those which are NS
- In **response models**, the interest is on which predictors are associated with the response
 - The minimal (null or baseline) model is the model of joint independence of the response (say, A) from all predictors, [A] [B C D . . .]
 - Associations among the predictors are fitted exactly (not analyzed)
 - Similar to regression, where predictors can be arbitrarily correlated
 - e.g., Berkeley data: fit the baseline model [Admit] [Gender Dept]
 - lack-of-fit $\implies$ associations [Admit Gender] and/or [Admit Dept]

Goodness of fit tests

As noted earlier, **overall** goodness of fit of a **specified** model may be tested by the likelihood ratio G^2 , or the Pearson X^2 ,

$$G^2 = 2 \sum_i n_i \log \left(\frac{n_i}{\hat{m}_i} \right) \quad X^2 = \sum_i \frac{(n_i - \hat{m}_i)^2}{\hat{m}_i} ,$$

with residual degrees of freedom $\nu = \# \text{ cells} - \# \text{ estimated parameters}$.

- These measure the **lack of fit** of a given model— a large value $\implies$ a poor model
- Both are distributed as $\chi^2(\nu)$ (in large samples: all $\hat{m}_i > 5$)
- $\mathcal{E}(\chi^2(\nu)) = \nu$, so G^2/ν (or X^2/ν) measures lack of fit per degree of freedom (**overdispersion**)
- But: how to compare or test competing models?

Nested models & ANOVA-type tests

Two models, M_1 and M_2 are **nested** when one (say, M_2) is a special case of the other

- Model M_2 (w/ v_2 df) fits a subset of the parameters of M_1 (w/ v_1 df)
- M_2 is more **restrictive** – cannot fit better than M_1 : $G^2(M_2) \geq G^2(M_1)$
- The least restrictive model is the **saturated** model [ABC ...], w/ $G^2 = 0$

Therefore, we can test the **difference in G^2** as a specific test of the added restrictions in M_2 compared to M_1 .

- This test has a χ^2 distribution with $df = v_2 - v_1$

$$\begin{aligned}\Delta G^2 \equiv G^2(M_2 | M_1) &= G^2(M_2) - G^2(M_1) \\ &= 2 \sum n_i \log(\hat{m}_{i1} / \hat{m}_{i2})\end{aligned}\tag{7}$$

Example: Berkeley admissions

For the UC Berkeley data, with table variables [A]dmit, [D]ept and [G]ender the following models form a nested chain

$$[A][D][G] \subset [A][DG] \subset [AD][AG][DG] \subset [ADG]$$

Table: Hierarchical G^2 tests for loglinear models fit to the UC Berkeley data

Type	LLM terms	G^2	df	$\Delta(G^2)$	$\Delta(df)$	$\Pr(> \Delta(G^2))$
Mutual ind	[A][D][G]	2097.67	16			
Joint	[A][DG]	877.06	11	1220.62	5	0.0000 ✓
All 2-way	[AD][AG][DG]	20.20	5	1128.70	5	0.0000 ✓
Saturated	[ADG]	0.0	0	20.20	5	0.0011 ✓

- Only testing the decrease in G^2 from one model to the next
- Here, each model is significantly better than the previous
- Joint vs. all two-way: Does Admit depend on Dept and/or Gender?
- Absolute fit of all 2-way model is not terrible. Investigate this further!

Fitting these in R

`loglm()` - data in contingency table form (MASS package)

```
1 data(UCBAdmissions)
2   ## conditional independence (AD, DG) in Berkeley data
3 mod.1 <- loglm(~ (Admit + Gender) * Dept, data=UCBAdmissions)
4   ## all two-way model (AD, DG, AG)
5 mod.2 <- loglm(~ (Admit + Gender + Dept)^2, data=UCBAdmissions)
```

`glm()` - data in frequency form

```
1 berkeley <- as.data.frame(UCBAdmissions)
2 mod.3 <- glm(Freq ~ (Admit + Gender) * Dept, data=berkeley,
3             family='poisson')
```

- `loglm()` simpler for nominal variables
- `glm()` allows a wider class of models and quantitative predictors (covariates)
- `gnm()` fits models for structured association and generalized *non-linear* models
- `vcdExtra` package provides visualizations for all.

Example: Berkeley admissions

Fit the model of mutual independence, using loglm()

```
> berk.loglm0 <- loglm(~ Admit + Dept + Gender, data=UCBAdmissions)
> berk.loglm0
Call:
loglm(formula = ~Admit + Dept + Gender, data = UCBAdmissions)

Statistics:

```

	X^2	df	P(> X^2)
Likelihood Ratio	2097.7	16	0
Pearson	2000.3	16	0

Conditional independence [AD] [AG]

```
> berk.loglm1 <- loglm(~ Admit * (Dept + Gender), data=UCBAdmissions)
> berk.loglm1
Call:
loglm(formula = ~Admit * (Dept + Gender), data = UCBAdmissions)

Statistics:

```

	X^2	df	P(> X^2)
Likelihood Ratio	1148.9	10	0
Pearson	1015.7	10	0

Conditional independence, [AD] [AG]

```
> berk.loglm2 <- loglm(~ Admit + (Dept * Gender), data=UCBAdmissions)
```

```
> berk.loglm2
```

```
Call:
```

```
loglm(formula = ~Admit + (Dept * Gender), data = UCBAdmissions)
```

```
Statistics:
```

	X^2	df	P(> X^2)
Likelihood Ratio	877.06	11	0
Pearson	797.70	11	0

All two-way model, [AD] [AG] [DG]

```
> berk.loglm3 <-loglm(~(Admit+Dept+Gender)^2, data=UCBAdmissions)
```

```
> berk.loglm3
```

```
Call:
```

```
loglm(formula = ~(Admit + Dept + Gender)^2, data = UCBAdmissions)
```

```
Statistics:
```

	X^2	df	P(> X^2)
Likelihood Ratio	20.204	5	0.0011441
Pearson	18.823	5	0.0020740

ANOVA tests

These are nested. Compare with **anova()**

```
> aov1 <- anova(berk.loglm0, berk.loglm1, berk.loglm3, test="Chisq")
> aov1
LR tests for hierarchical log-linear models
```

Model 1:

~Admit + Dept + Gender

Model 2:

~Admit * (Dept + Gender)

Model 3:

~(Admit + Dept + Gender)^2

	Deviance	df	Delta(Dev)	Delta(df)	P(> Delta(Dev))	
Model 1	2097.671	16				
Model 2	1148.901	10	948.770	6	0.00000	✓
Model 3	20.204	5	1128.697	5	0.00000	✓
Saturated	0.000	0	20.204	5	0.00114	✓

These are tests of **relative fit**, $\Delta G^2 = G^2(M_i | M_{i-1})$

LRstats: AIC & BIC

`vcdExtra::LRstats()` gives one-line summaries of a collection of models
These are tests of **absolute** goodness of fit

```
> LRstats(berk.loglm0, berk.loglm1, berk.loglm2, berk.loglm3)
```

Likelihood summary table:

	AIC	BIC	LR	Chisq	Df	Pr(>Chisq)	
berk.loglm0	2273	2282		2098	16	<2e-16	***
berk.loglm1	1336	1352		1149	10	<2e-16	***
berk.loglm2	1062	1077		877	11	<2e-16	***
berk.loglm3	217	240		20	5	0.0011	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- AIC and BIC are GOF measures adjusted for model **parsimony**
- Not not significance tests, but **smaller is better**
- Also apply to **non-nested** models

$$AIC = G^2 + 2 \times \# \text{ parameters}$$

$$BIC = G^2 + 2 \log(n) \times \# \text{ parameters}$$

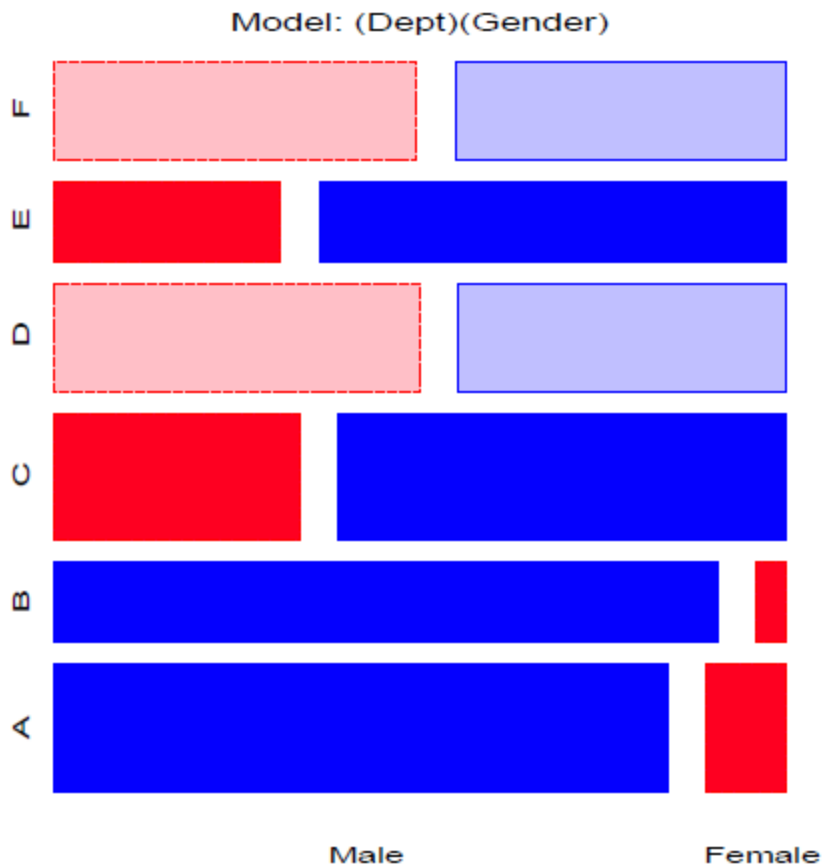
Mosaic displays: Visual fitting

- Each mosaic shows:
 - The DATA – size of tiles
 - (some) marginal frequencies – initial splits (visual grouping)
 - RESIDUALS (shading) – what associations have been omitted?
- Visual fitting
 - Start with a simple model: mutual independence or joint independence for response models
 - Pattern of residuals: suggest a better model → smaller residuals
 - Add terms: → smaller residuals, less shading: “cleaning the mosaic”
 - Good fitting model will have mostly unshaded tiles

Mosaic displays: Predictor variables

Berkeley data: Departments $\times$ Gender (ignoring Admit):

- Did departments differ in the total number of applicants?
- Did men and women apply differentially to departments?



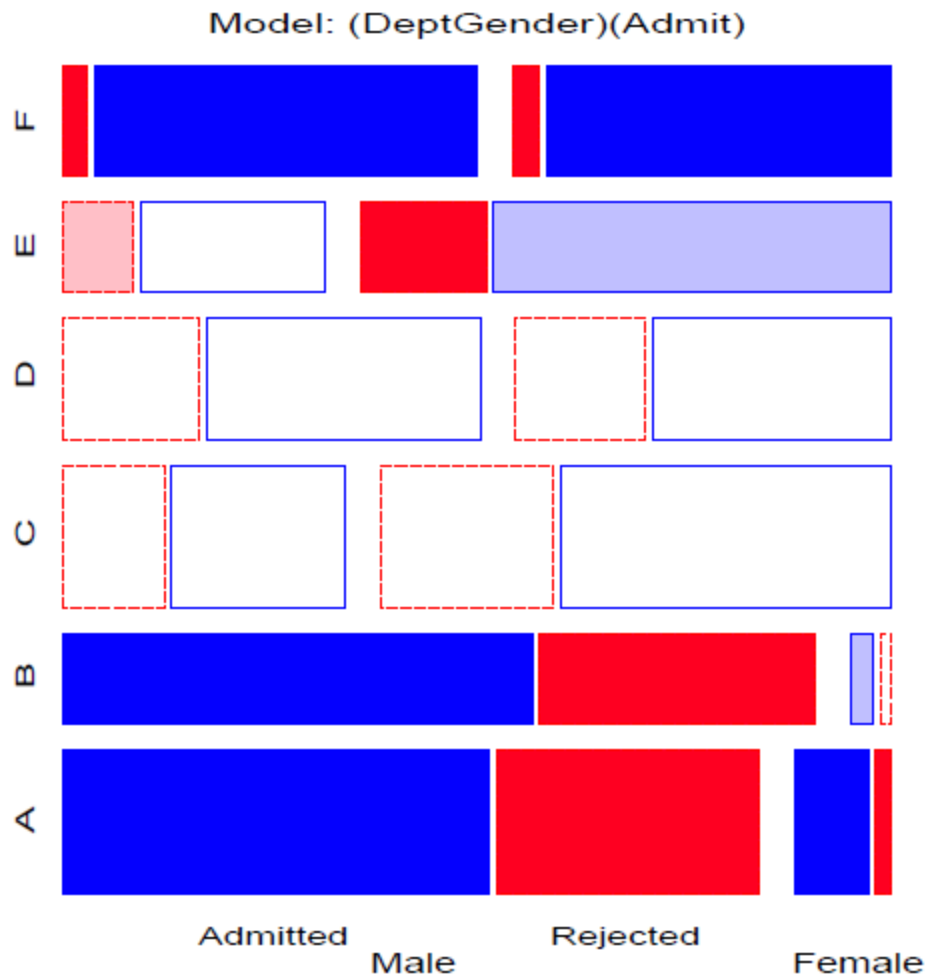
In response models, the mosaic of the predictors gives a graphic summary of **background** variables

- Model [Dept] [Gender]: $G^2_{(5)} = 1220.6$.
- *Note*: Departments ordered A–F by overall rate of admission.
- Men more likely to apply to departments A,B; women more likely in depts C–F

Baseline response model

For the Berkeley data, start with the model of [joint independence](#), [A][DG]

Fits badly: $G^2_{(11)} = 877.1$



This is the [null](#), or [baseline](#) model when Admit is the response variable.

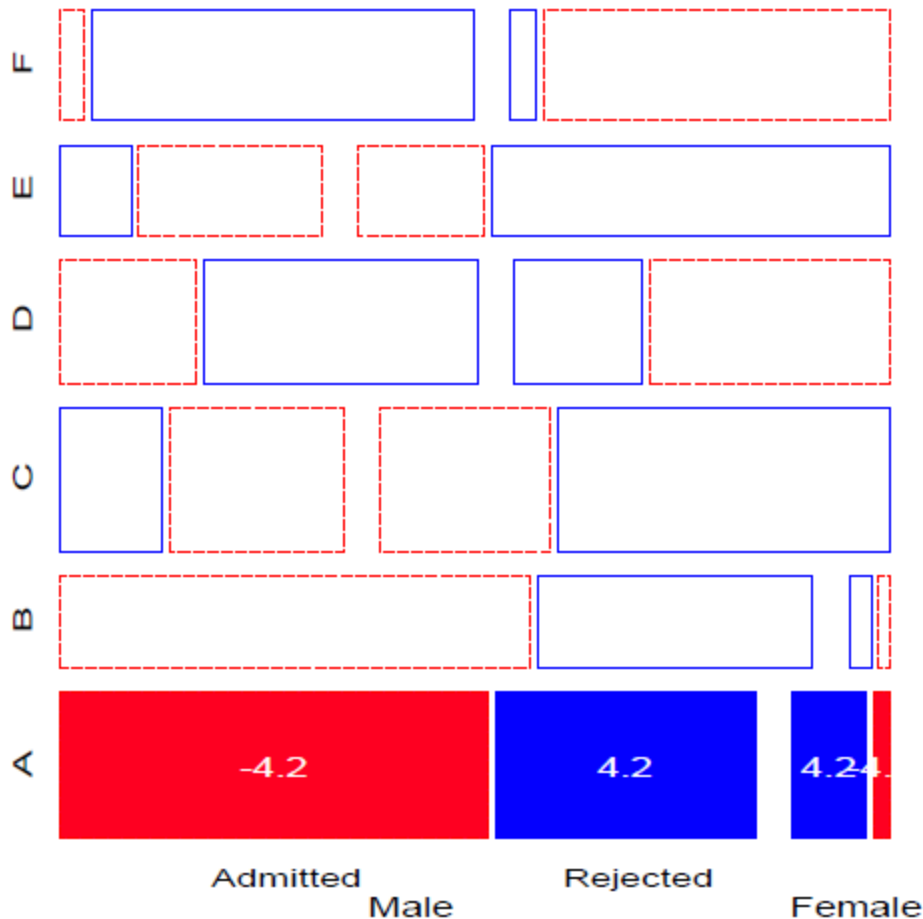
Allows/fits assoc. of [Dept Gender], not shown in shading

Remaining shading suggests:
[AD] : Admit varies w/ Dept
[AG] : Admit varies w/ Gender

Add an association term...

Conditional independence, [AD] [DG]:

Model: (DeptGender)(DeptAdmit)

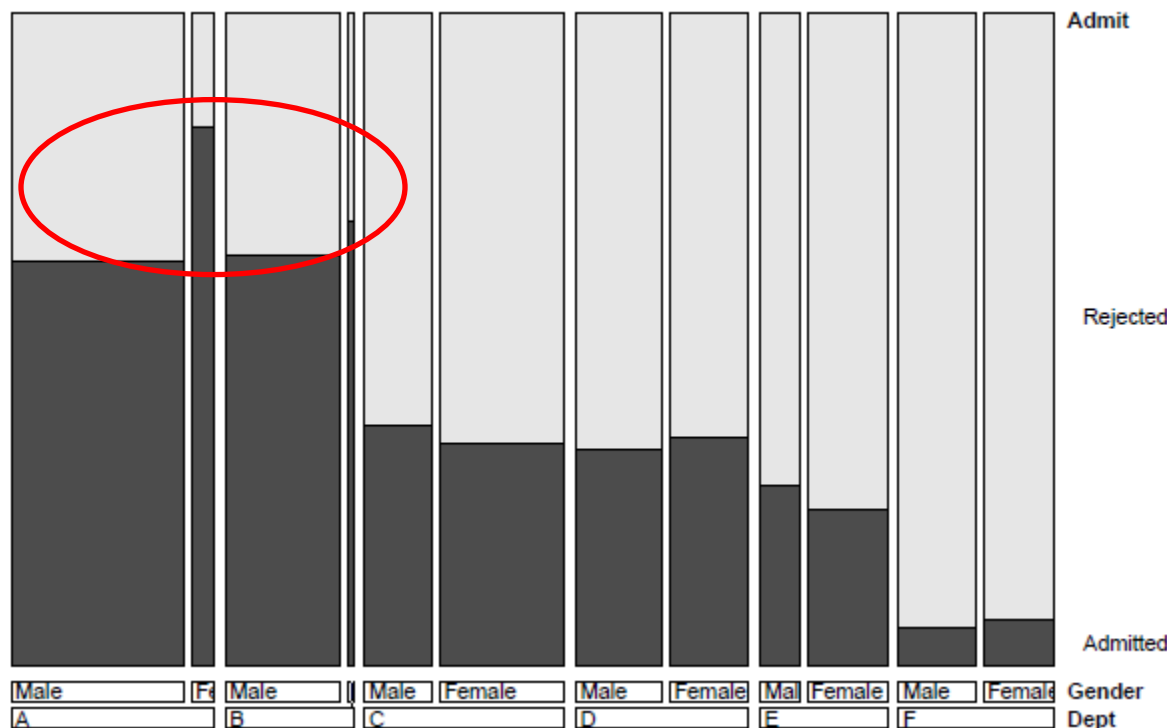


- E.g., Add [Admit Dept] association → Conditional independence:
 - Fits poorly: ($G^2_{(6)} = 21.74$)
 - But, only in Department A!
- GLM approach allows fitting a special term for Dept. A
- Note: These displays use *standardized residuals*: better statistical properties.

Double decker plots

Visualize dependence of one **response** variable (typically binary) on combinations of predictors
Formally: mosaic plots with vertical splits for all predictors, **highlighting** the response by shading

```
doubledecker(Admit ~ Dept + Gender, data = UCBAmissions[2:1, ,])
```



An exploratory plot

Highlights the M-F
diff^{ce} in Admit for
Dept A

DDAR Fig 5.34, p 211



Survival on the *Titanic*

An epic data set, revealed with loglinear models

4-way tables: Survival on the *Titanic*

Data on the fate of passengers & crew on the HMS *Titanic*: a $4 \times 2 \times 2 \times 2$ table

```
> data(Titanic, package="datasets")
> str(Titanic)
'table' num [1:4, 1:2, 1:2, 1:2] 0 0 35 0 0 0 17 0 118 154 ...
- attr(*, "dimnames")=List of 4
 ..$ Class      : chr [1:4] "1st" "2nd" "3rd" "Crew"
 ..$ Sex         : chr [1:2] "Male" "Female"
 ..$ Age         : chr [1:2] "Child" "Adult"
 ..$ Survived: chr [1:2] "No" "Yes"
```

What proportion survived? Ans: $711/2201 = 32.3\%$

```
> addmargins(margin.table(Titanic, 4))
Survived
  No   Yes  Sum
1490  711 2201
> margin.table(Titanic, 4) / sum(Titanic)
Survived
  No   Yes
0.677 0.323
```

Zero cells

```
> structable(Titanic)
```

		Sex		Male		Female	
		Survived		No	Yes	No	Yes
Class	Age						
1st	Child			0	5	0	1
	Adult			118	57	4	140
2nd	Child			0	11	0	13
	Adult			154	14	13	80
3rd	Child			35	13	17	14
	Adult			387	75	89	76
Crew	Child			0	0	0	0
	Adult			670	192	3	20

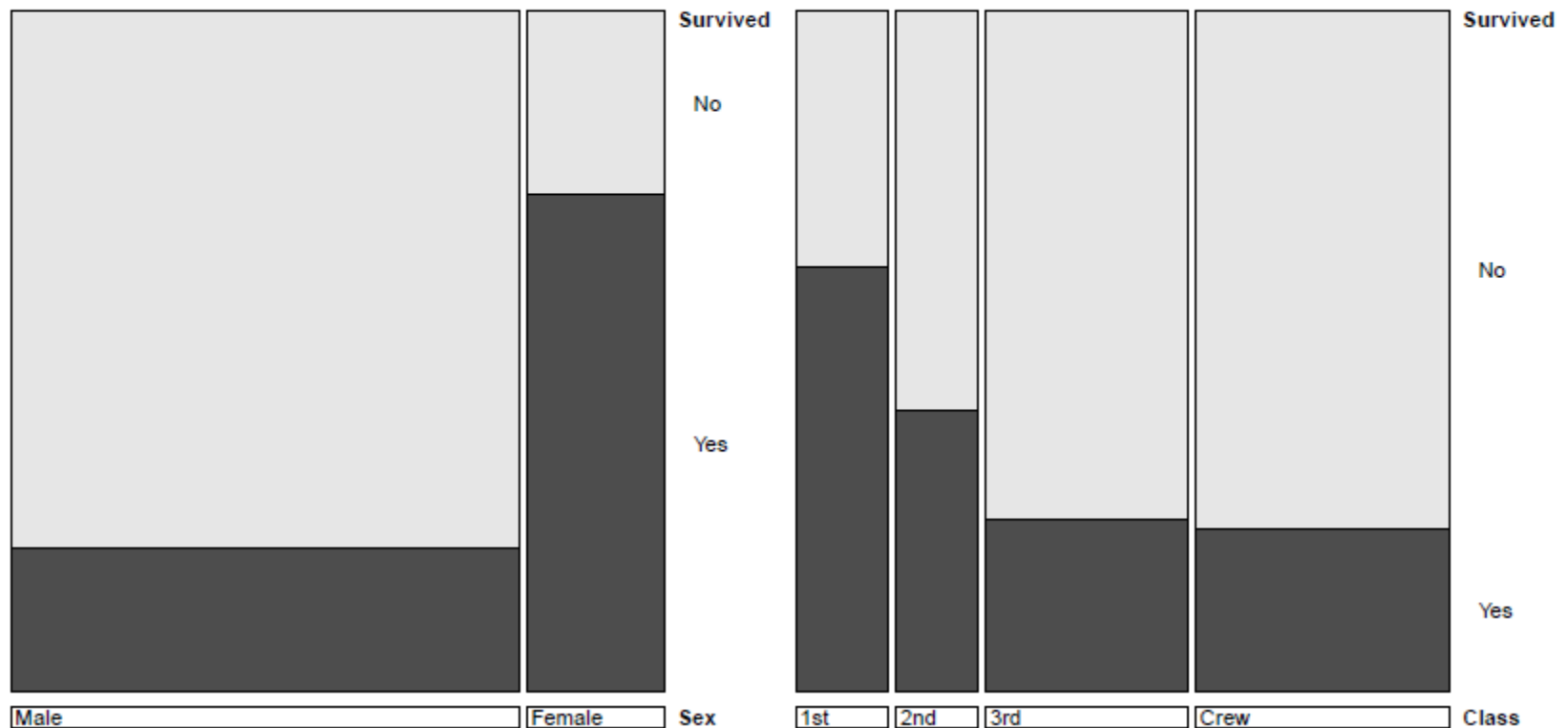
Two types of zero cells:

- **Structural zeros**: could not occur (children in crew)
- **Sampling zeros**: did not happen to occur (children in 1st & 2nd who died)
- Beware: zeros can cause problems:
 - Loss of df
 - $0/0 \rightarrow \text{NaN}$ in χ^2 tests

Exploratory plots

One-way doubledecker plots against survival show what might be expected:

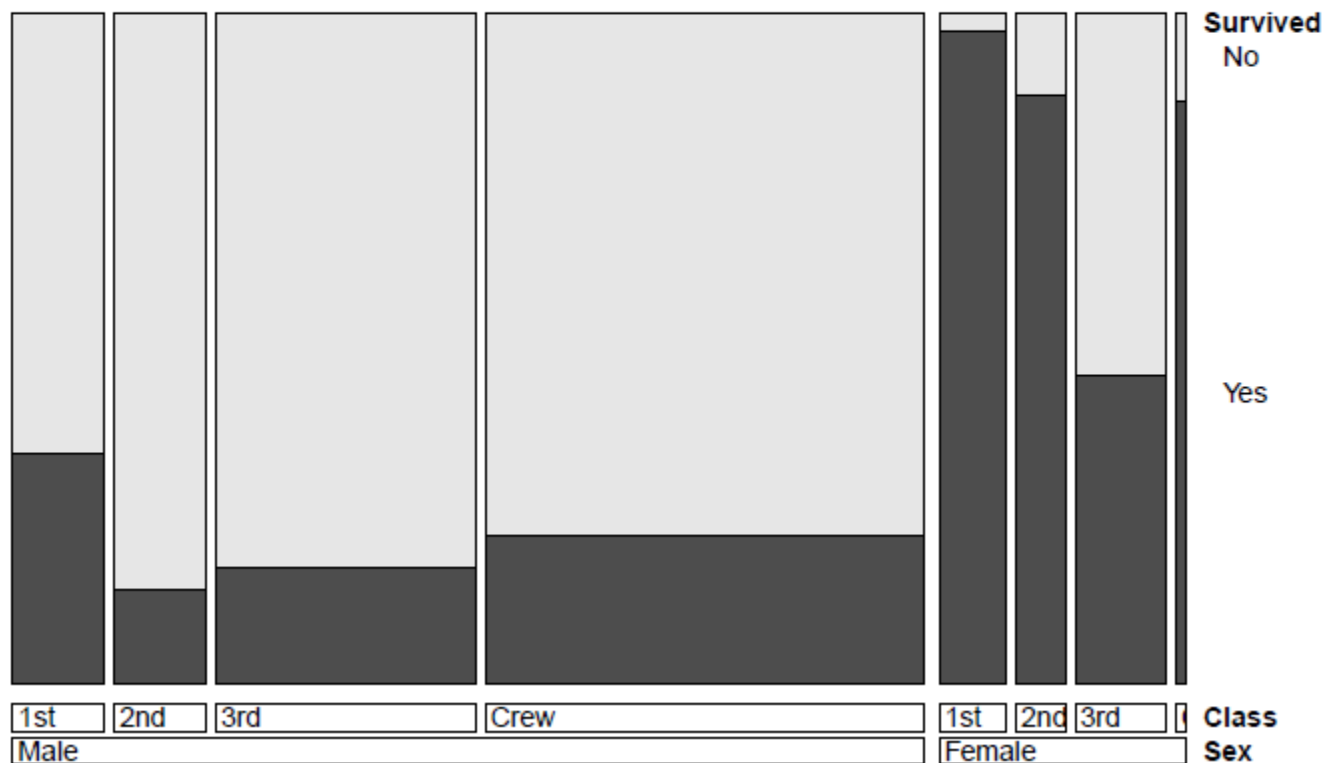
```
doubledecker(Survived ~ Sex, data=Titanic)
doubledecker(Survived ~ Class, data=Titanic)
```



Exploratory plots

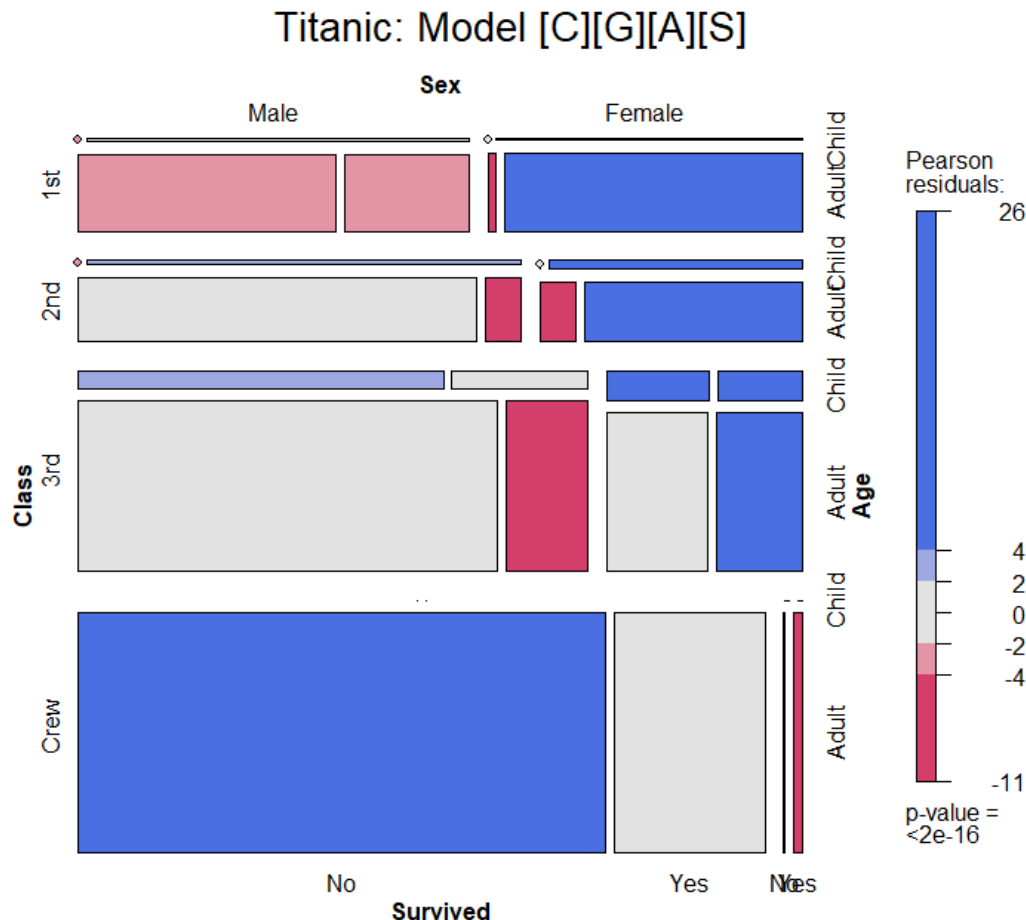
Two-way doubledecker plot against survival shows different effects of `Class` for men and women:

```
doubledecker(Survived ~ Sex + Class, data=Titanic)
```



Fitting & visualizing models

```
mod0 <- loglm(~ 1 + 2 + 3 + 4, data=Titanic)
mosaic(mod0, main="Titanic: Model [C][G][A][S]")
```



In the model formulas, I'm using variable numbers 1-4 for **C**lass, **G**ender, **A**ge and **S**urvived

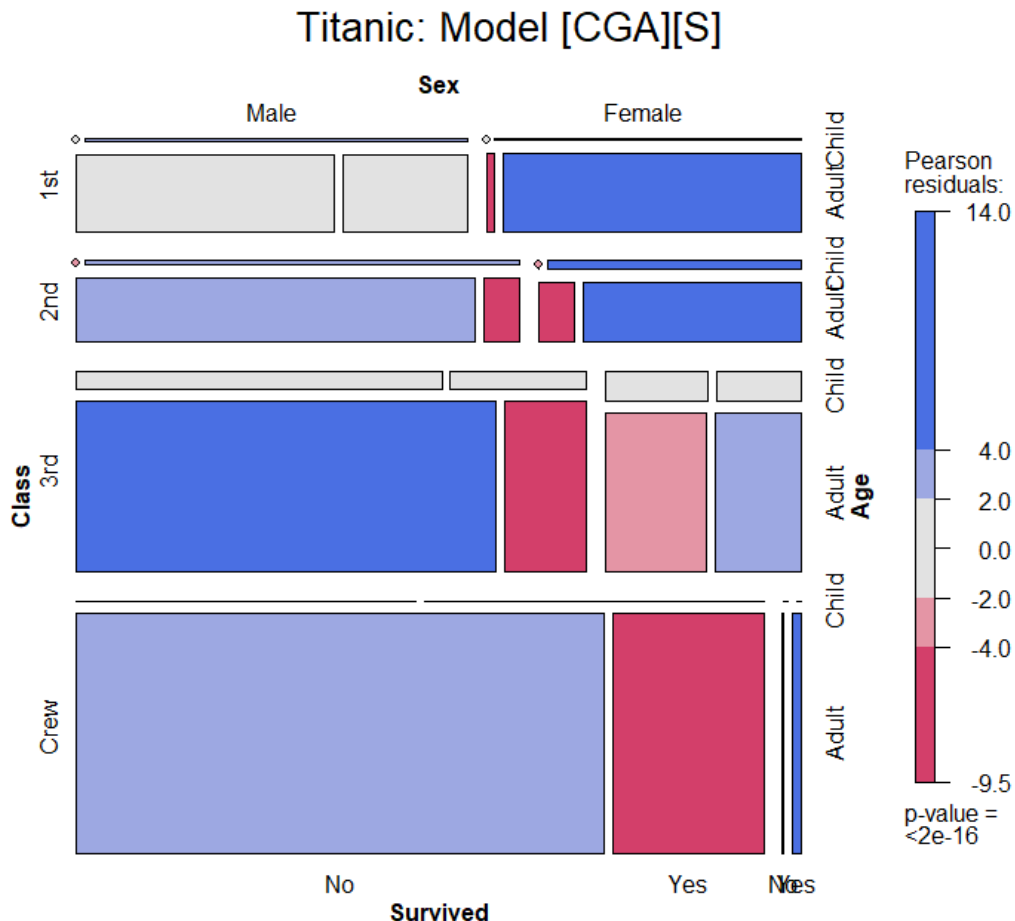
The **independence** model serves only as a background for the total associations in the table

Let's clean this mosaic!!

Note the scale of residuals:
+26 -- -11

Baseline model for Survived

```
mod1 <- loglm(~ 1*2*3 + 4, data=Titanic)
mosaic(mod1, main="Titanic: Model [CGA][S]")
```



With **S** as response, the **baseline** model includes all associations among [CGA]

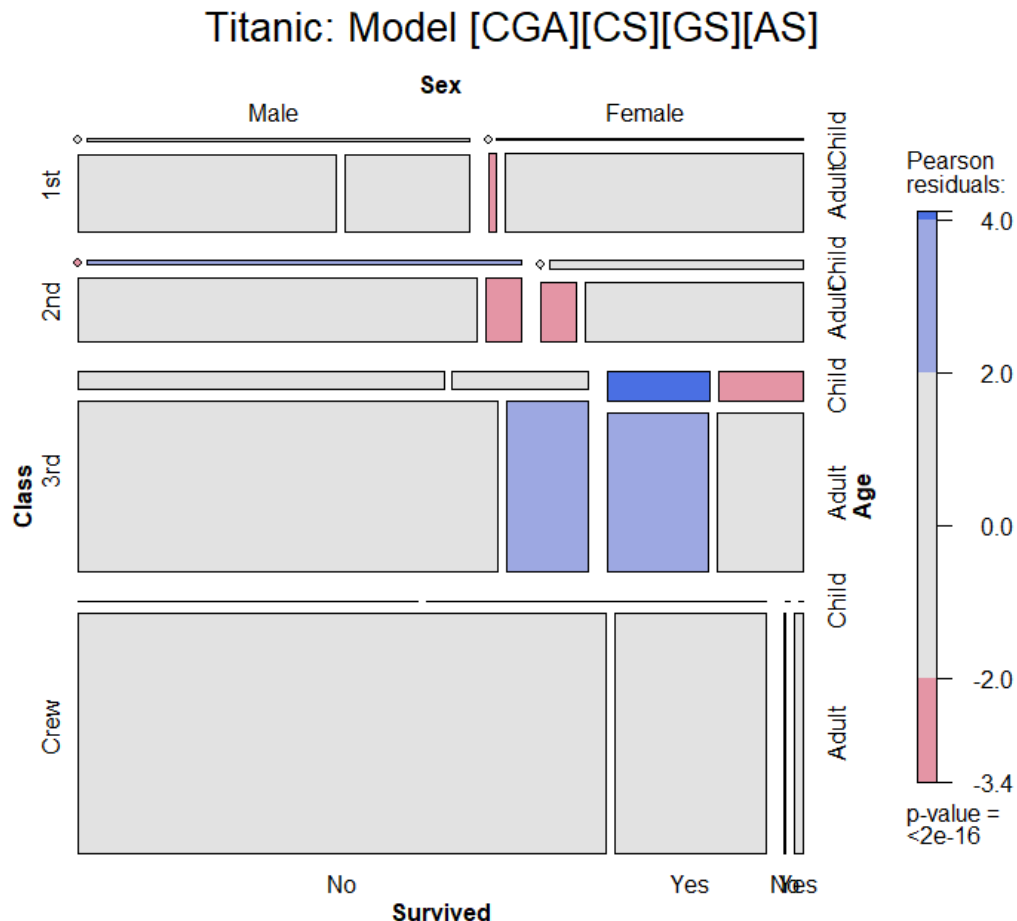
But this model asserts survival is independent of all of these

$G^2_{(15)} = 671.96$, a very poor fit

Pattern of residuals shows all associations with Surv

Adding associations: Main effects

```
mod2 <- loglm(~ 1*2*3 + (1+2+3)*4, data=Titanic)
mosaic(mod2, main="Titanic: Model [CGA][CS][GS][AS]")
```

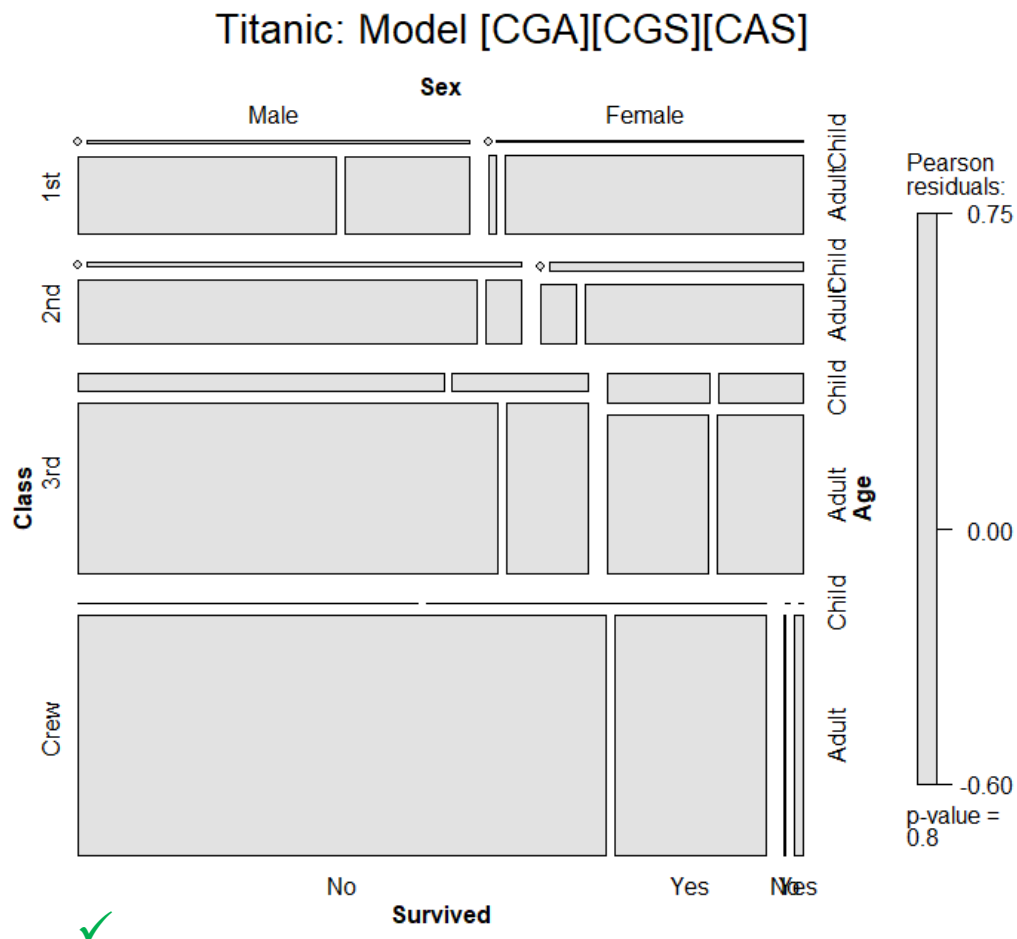


- This model allows associations of each of C, G, A with Survived
- $G^2(10) = 112.57$, still not good
- Pattern of residuals suggests 2-way interactions (3-way terms):
- “Women & children first”: suggests a term [GAS]
- Allow interactions of Class with Gender [CGS] and Class with Age [CAS]

Final model

```
mod3 <- loglm(~ 1*2*3 + (1*2)*4 + (1*3)*4, data=Titanic)
mosaic(mod3, main="Titanic: Model [CGA][CGS][CAS]")
```

Now I understand



Nice & clean!

$$G^2_{(4)} = 1.69, p=0.79$$

Before accepting this,
should compare models,
and consider

- parsimony
- model explanations

Comparing models

As usual, **anova()** give compact **relative** comparisons of a set of nested models

```
> anova(mod0, mod1, mod2, mod3)
LR tests for hierarchical log-linear models
```

Model 1:

~1 + 2 + 3 + 4

Model 2:

~1 * 2 * 3 + 4

Model 3:

~1 * 2 * 3 + (1 + 2 + 3) * 4

Model 4:

~1 * 2 * 3 + (1 * 2) * 4 + (1 * 3) * 4

	Deviance	df	Delta(Dev)	Delta(df)	P(> Delta(Dev))
Model 1	1243.66	25			
Model 2	671.96	15	571.70	10	0.000
Model 3	112.57	10	559.40	5	0.000
Model 4	1.69	4	110.88	6	0.000
Saturated	0.00	0	1.69	4	0.793

Q: Please help me interpret these results 😊

Comparing models

LRstats() gives **absolute** GOF tests; also provides AIC, BIC stats: model parsimony

```
> LRstats(mod0, mod1, mod2, mod3)
Likelihood summary table:
      AIC  BIC LR Chisq Df Pr(>Chisq)
mod0 1385 1395      1244 25      <2e-16 ***
mod1  833  858      672 15      <2e-16 ***
mod2  284  316      113 10      <2e-16 ***
mod3  185  226         2  4        0.79
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

mod3 [CGA] [CGS] [CAS] wins!

- Acceptable G^2
- Looks best by AIC & BIC



Model interpretation



Recall that the goal of analysis is to tell a story

- Greatest impact: lower class → decreased survival, regardless of Gender & Age
- Differences in survival by Class were **moderated** by both Gender & Age
 - Term [CGS]: Women in 3rd class did not have an advantage, while men in 1st class did vs. other classes
 - Term [CAS]: No children in 1st or 2nd class died, but nearly 2/3 in 3rd class did
- Summary:
 - Not so much “women & children first”, rather
 - Women & children, ordered by class, and 1st class men!

Historical note

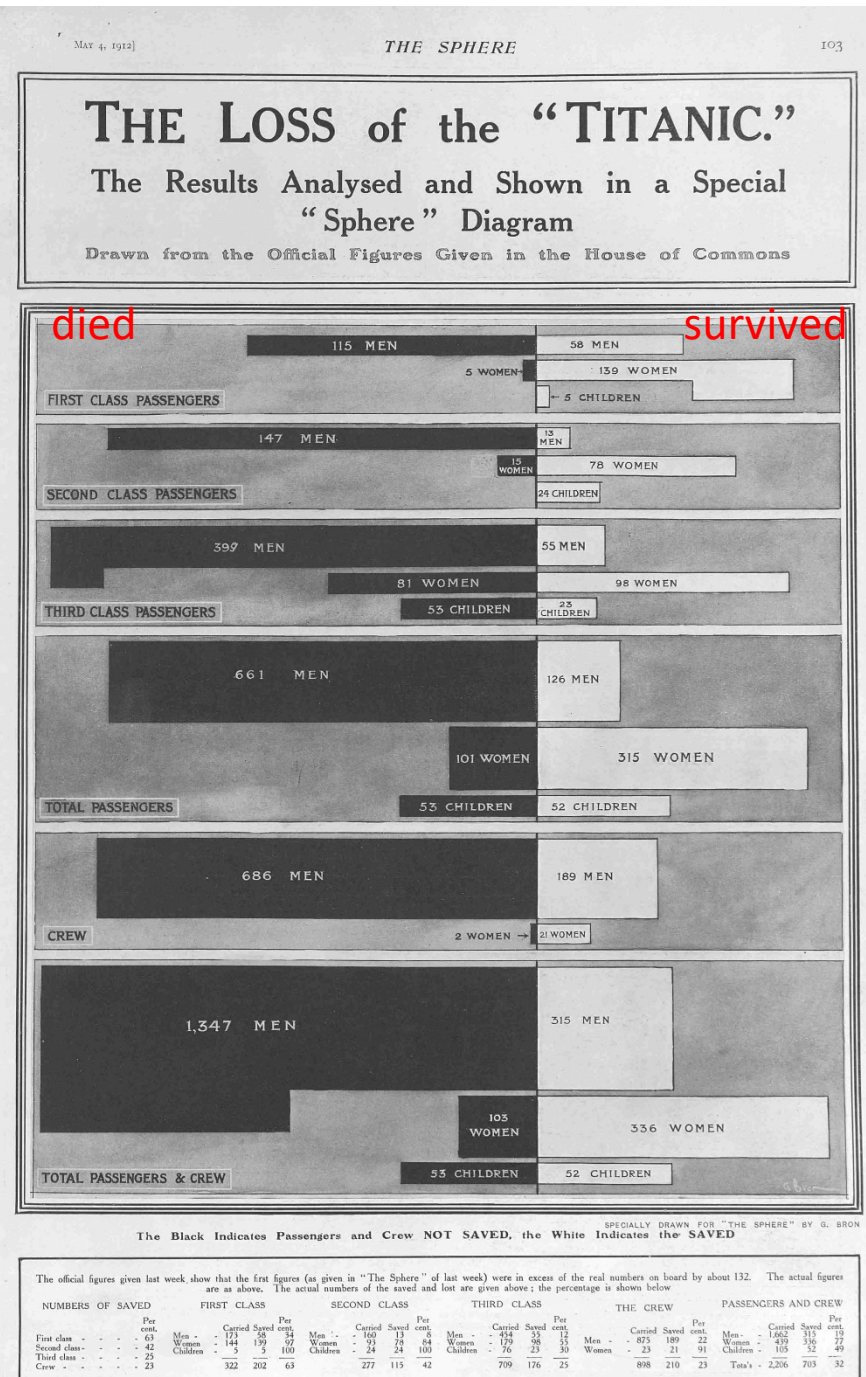
The *Titanic* sank on Apr. 15, 1912

On May 4, the technical illustrator, G. Bron published this graph in *The Sphere*, a popular magazine.

He used a remarkably modern graph to show the differences in survival by class, gender & age

Read the story:

Friendly, Symanzik, Onder, [Visualizing the Titanic Disaster](#), *Significance*, Feb., 2019



A woman with long blonde hair, wearing a light blue sequined dress, is singing into a microphone on a stage. She is smiling and looking towards the camera. The background is dark with some stage lights visible.

Sequential plots & models

Do you want another one?

Sequential plots & models

- Mosaic for an n-way table → hierarchical decomposition of association
- Joint cell probabilities are decomposed as:

$$p_{ijk\ell\ldots} = \underbrace{p_i \times p_{j|i}}_{\{v_1 v_2\}} \times p_{k|ij} \times p_{\ell|ijk} \times \cdots \times p_{n|ijk\ldots}$$

- First 2 terms: → mosaic for $\{v_1, v_2\}$
 - First 3 terms: → mosaic for $\{v_1, v_2, v_3\}$
 - ... and so on
- Roughly analogous to **sequential** fitting in regression: $X_1 ; X_2 | X_1 ; X_3 | X_1, X_2$
- Order of variables matters for interpretation
 - Mosaics: 1st split: easiest to see the marginal proportions
 - Mosaics: 2nd variable seen as conditional proportions, given the 1st

Sequential plots & models

- Sequential models of joint independence
 - Give an **additive** decomposition of **total** association – mutual independence $[v_1][v_2] \dots [v_p]$

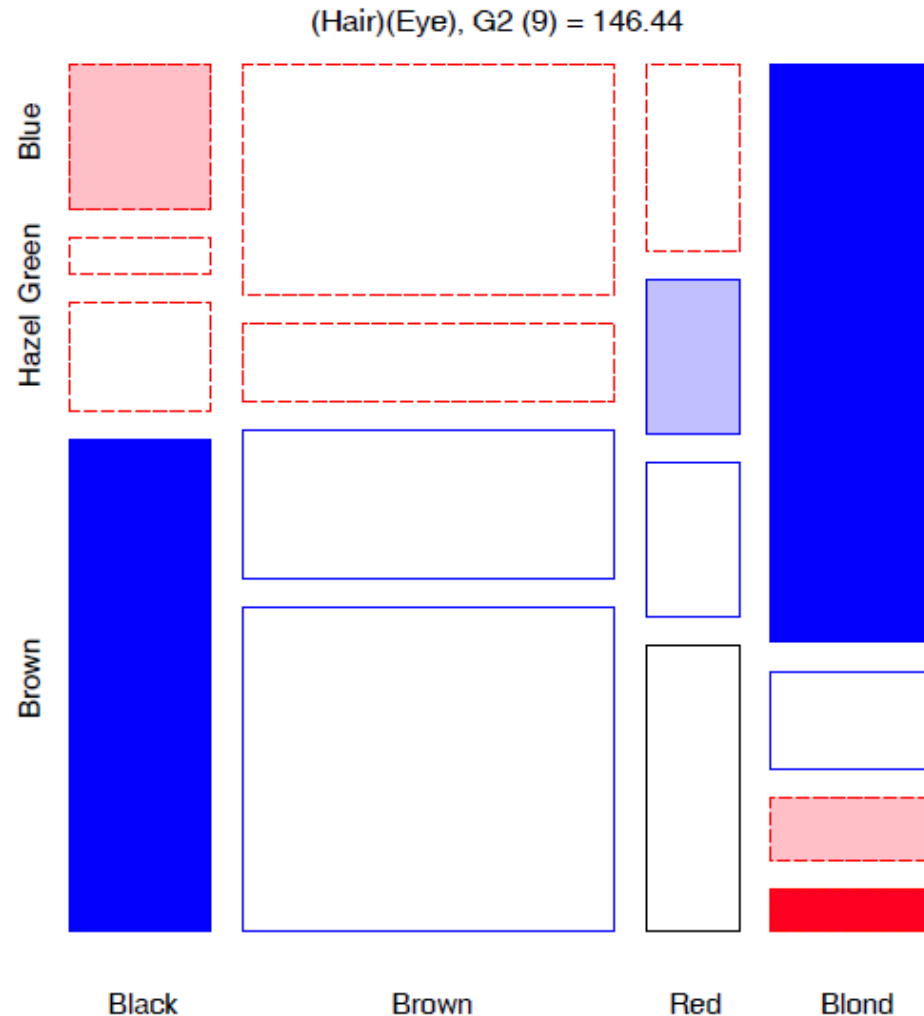
$$G^2_{[v_1][v_2]\dots[v_p]} = G^2_{[v_1][v_2]} + G^2_{[v_1 v_2][v_3]} + G^2_{[v_1 v_2 v_3][v_4]} + \dots + G^2_{[v_1 \dots v_{p-1}][v_p]}$$

- E.g., for Hair Eye color data

Model	Model symbol	df	G^2
Marginal	[Hair] [Eye]	9	146.44
Joint	[Hair, Eye] [Sex]	15	19.86
Mutual	[Hair] [Eye] [Sex]	24	166.30

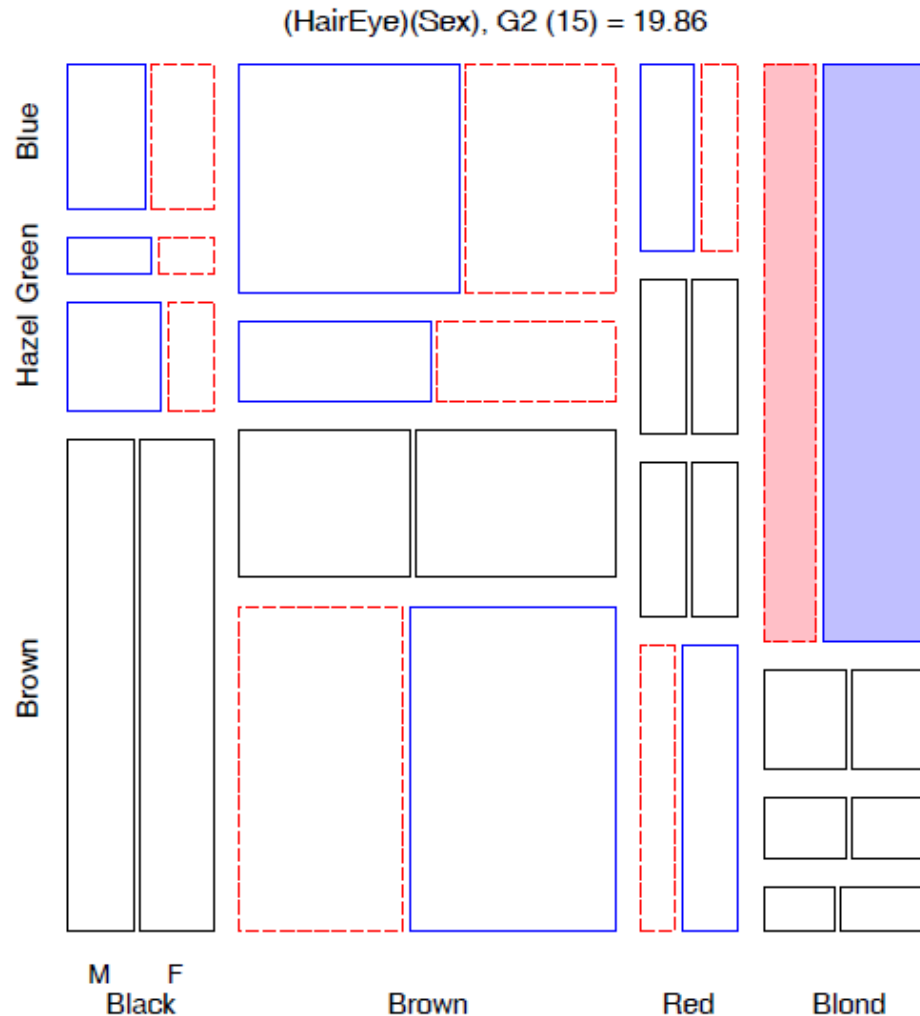
Sequential plots & models

Hair color × Eye color marginal table (ignoring Sex)



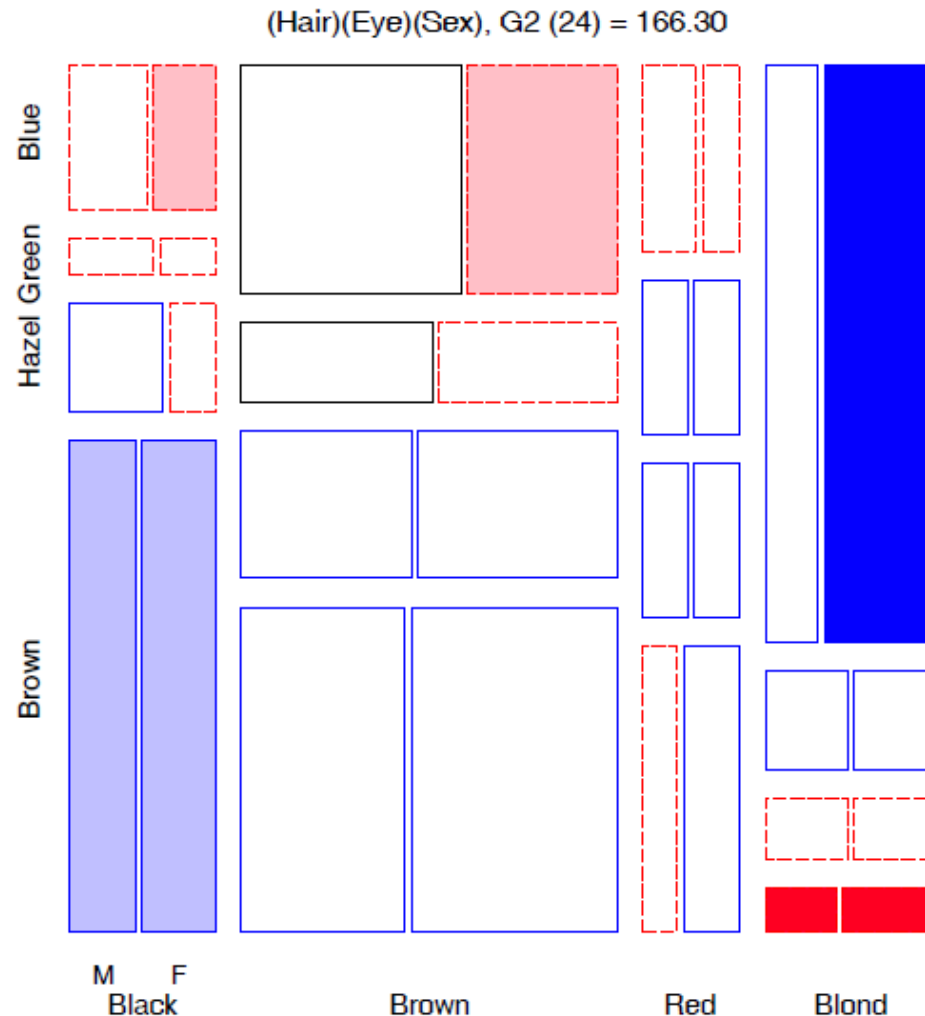
Sequential plots & models

3-way table, Joint independence model [Hair Eye][Sex]



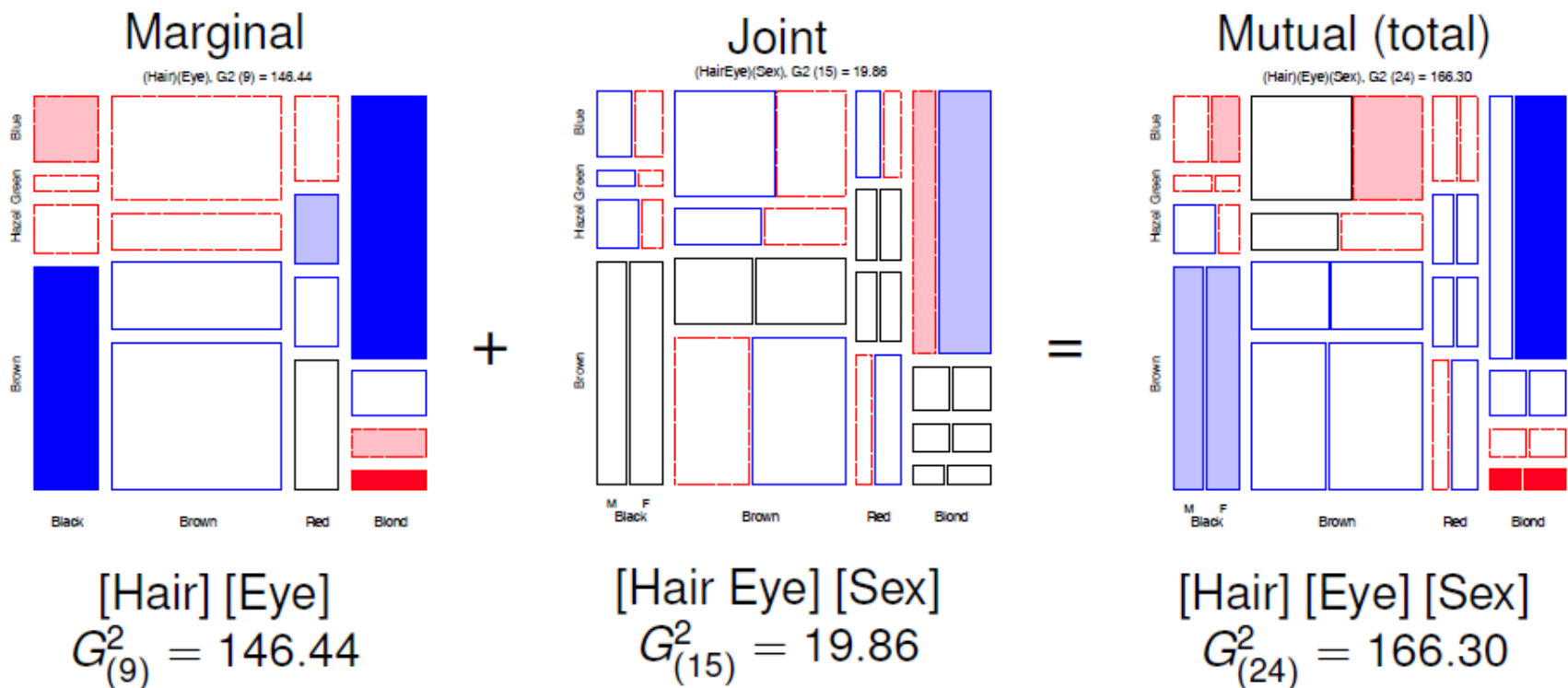
Sequential plots & models

3-way table, Mutual independence [Hair] [Eye][Sex]



Sequential plots & models

Putting these together:



TIP: `vcdExtra::seq_loglm()` generates a variety of sequential models

Sequential models: Applications

Response models

- When one variable, R , is a response and $E_1, E_2, \dots$ are explanatory, the baseline model is the model of joint independence, $[E_1, E_2, \dots][R]$
- Sequential mosaics then show the associations among the predictors
- The last mosaic shows all associations with R
- Better-fitting models will need to add associations of the form $[E_i R], [E_i E_j R] \dots$

Causal models

- Sometimes there is an assumed **causal ordering** of variables:

$$A \rightarrow B \rightarrow C \rightarrow D$$

- Each path of arrows: $A \rightarrow B$, $A \rightarrow B \rightarrow C$ is a sequential model of joint independence: $[A][B]$, $[AB][C]$, $[ABC][D]$.
- Testing these decomposes all joint probabilities

Example: Marital status, pre- & extra-marital sex

Thornes and Collard (1979) studied divorce patterns in relation to premarital and extramarital sex, a 2^4 table, **PreSex** in vcd ($G \times P \times E \times M$)

```
> data("PreSex", package="vcd")
> structable(Gender + PremaritalSex + ExtramaritalSex ~
             MaritalStatus, data = PreSex)
```

	Gender	Women				Men			
	PremaritalSex	Yes		No		Yes		No	
	ExtramaritalSex	Yes	No	Yes	No	Yes	No	Yes	No
MaritalStatus									
Divorced		17	54	36	214	28	60	17	68
Married		4	25	4	322	11	42	4	130

Submodels:

- ❖ $[G][P]$: Do men & women differ by pre-marital sex?
- ❖ $[GP][E]$: Given G & P, are there differences in extra-marital sex?
- ❖ $[GPE][M]$: Are there differences in divorce among the G, P, E groups?

Debugging color_table()

```
> color_table(PreSex, formula = Gender + PremaritalSex + ExtramaritalSex ~ MaritalStatus)
```

Shading based on residuals from model of complete independence, $X^2 = 270.14$, $df = 11$, $p = 0$

PreSex: Marital Status & Gender by Sexual Behavior									
	Gender_PremaritalSex_ExtramaritalSex								Total
	Women_Yes_Yes	Women_Yes_No	Women_No_Yes	Women_No_No	Men_Yes_Yes	Men_Yes_No	Men_No_Yes	Men_No_No	
Divorced	17	54	36	214	28	60	17	68	494
Married	4	25	4	322	11	42	4	130	542
Total	21	79	40	536	39	102	21	198	1036

```
> color_table(PreSex, formula = MaritalStatus + Gender ~ PremaritalSex + ExtramaritalSex)
```

PreSex: Marital Status & Gender by Sexual Behavior					
	MaritalStatus_Gender				Total
	Divorced_Women	Divorced_Men	Married_Women	Married_Men	
Yes_Yes	17	28	4	11	60
Yes_No	54	60	25	42	181
No_Yes	36	17	4	4	61
No_No	214	68	322	130	734
Total	321	173	355	187	1036

What's missing here?

- Rows & cols each have nested categories
- Need to reflect this in the labels, as is done in structable()

But: what cells stand out?

Example: Marital status, pre- & extra-marital sex

Order the table variables as $G \rightarrow P \rightarrow E \rightarrow M$

```
> names(dimnames(PreSex))      # table variable names
[1] "MaritalStatus"  "ExtramaritalSex" "PremaritalSex"  "Gender"

> PreSex <- aperm(PreSex, 4:1)  # order variables G, P, E, M
```

Fit each sequential model to the marginal sub-table. **vcdExtra::seq_loglm()** generates these models of **joint independence**

```
PreSex.mods <- seq_loglm(PreSex,
                        type="joint",
                        marginals = 2:4)

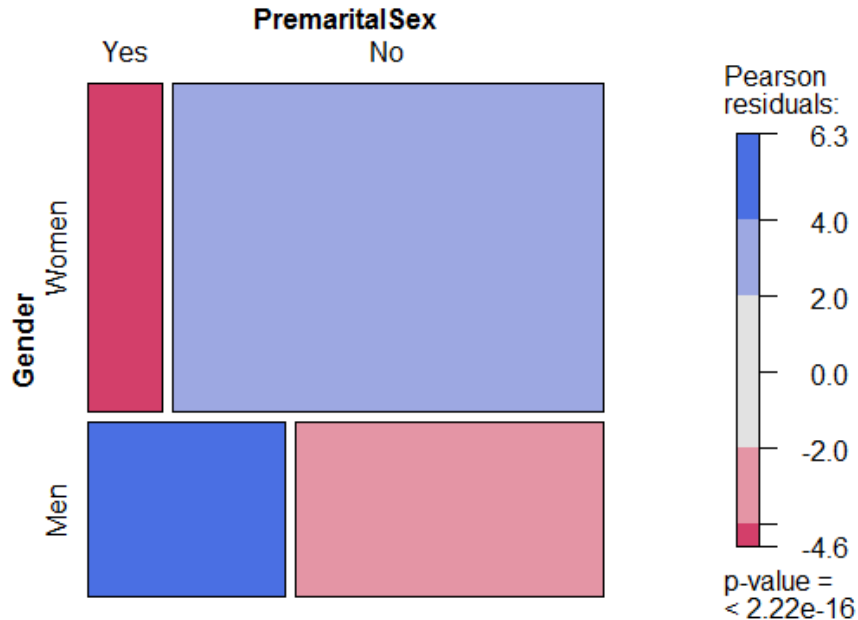
LRstats(PreSex.mods)
```

Model	df	G^2
[G] [P]	1	75.259
[GP] [E]	3	48.929
[GPE] [M]	7	107.956
[G] [P] [E] [M]	11	232.142

Mosaic plots

```
# (Gender Pre)
mosaic(margin.table(PreSex, 1:2), shade=TRUE,
       main = "Gender and Premarital Sex")
```

Gender and Premarital Sex

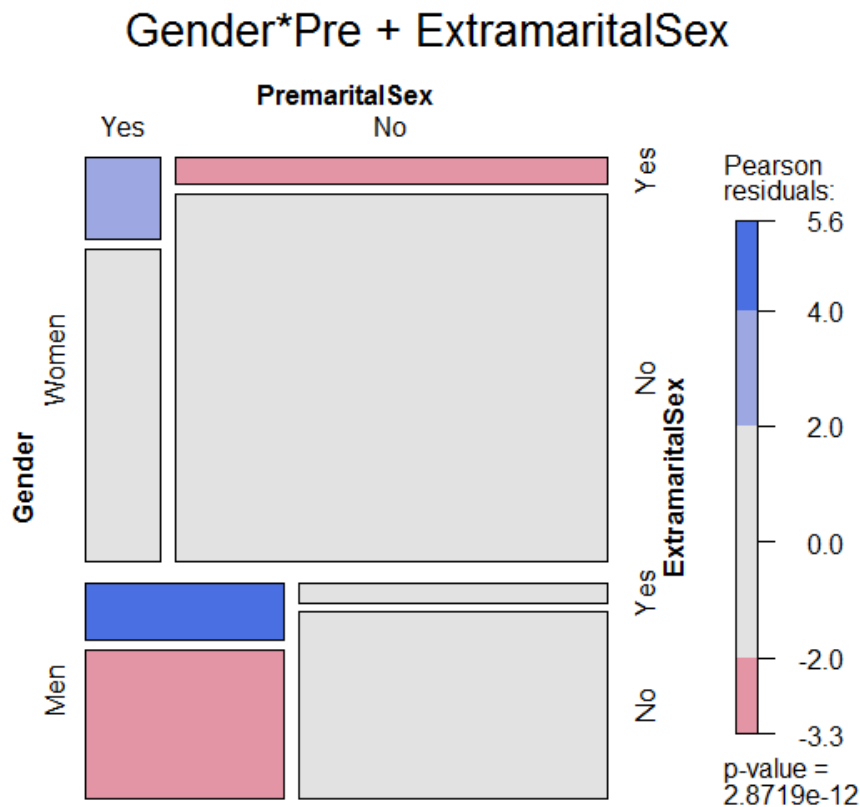


Twice as many women in this sample

Men far more likely to report pre-marital sex than women (odds ratio = 3.7)

Mosaic plots

```
# (Gender Pre) (Extra)
mosaic(margin.table(PreSex, 1:3),
       expected = ~Gender * PremaritalSex + ExtramaritalSex,
       main = "Gender*Pre + ExtramaritalSex")
```

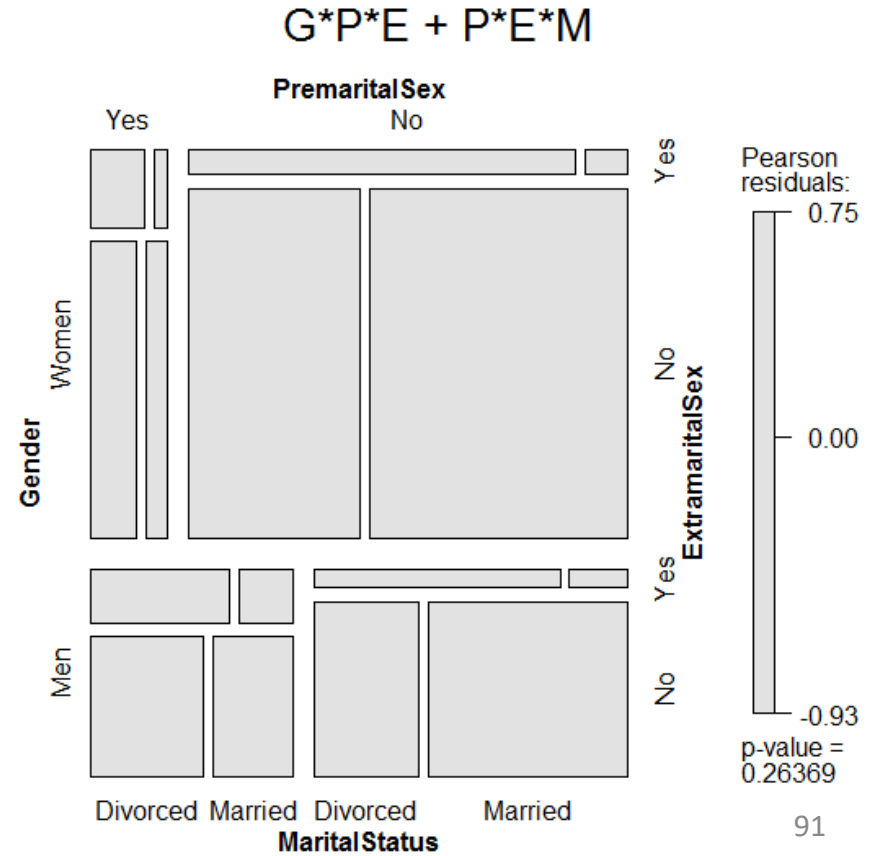
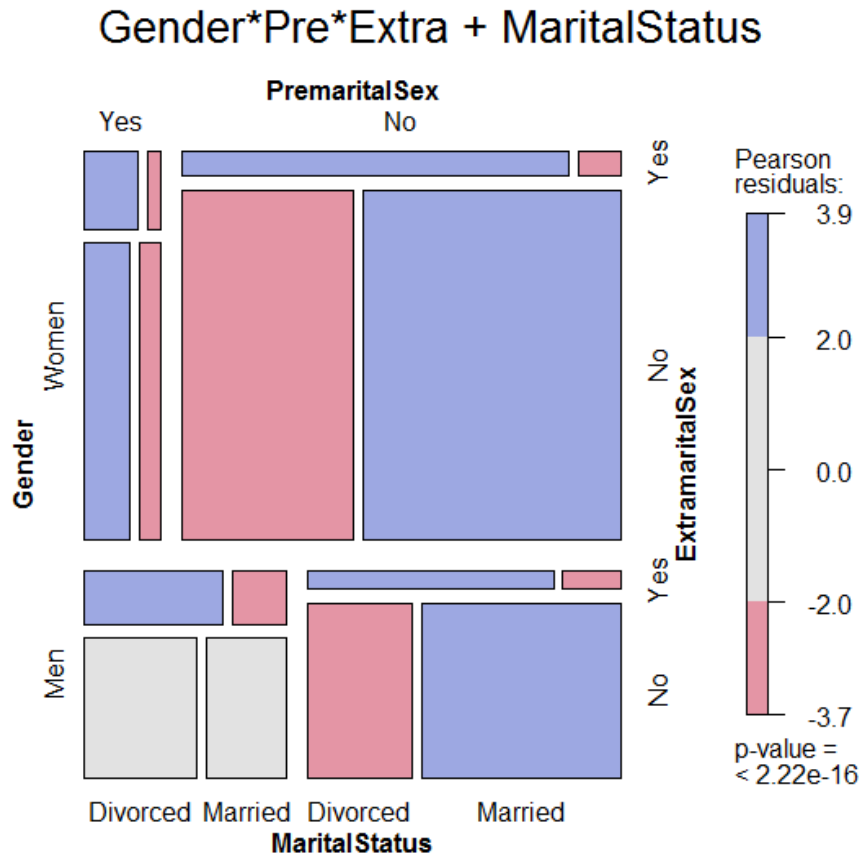


Men & women who reported Pre-
far more likely to report Extra- sex

Odds ratio of Extra- given Pre-
about the same for men & women
(3.61 vs. 3.56)

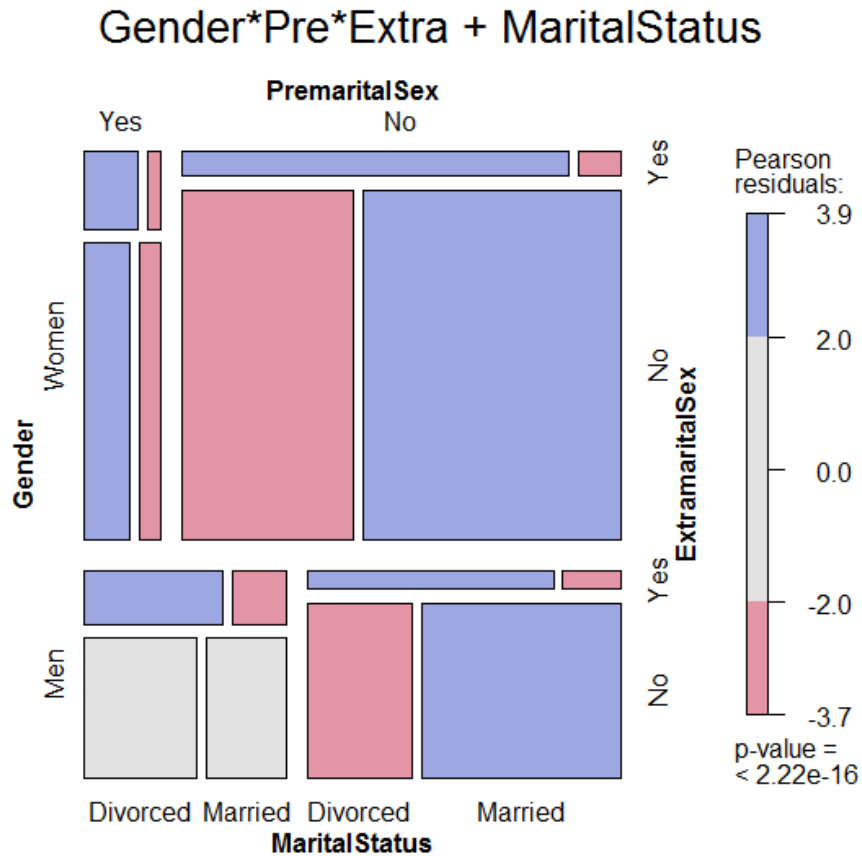
Mosaic plots: Full table

```
mosaic(PreSex,
      expected = ~Gender * PremaritalSex * ExtramaritalSex
      + MaritalStatus,
      main = "Gender*Pre*Extra + MaritalStatus")
# (GPE) (PEM)
mosaic(PreSex,
      expected = ~ Gender * PremaritalSex * ExtramaritalSex
      + MaritalStatus * PremaritalSex * ExtramaritalSex,
      main = "G*P*E + P*E*M")
```



Mosaic plots: Full table

```
mosaic(PreSex,  
       expected = ~Gender * PremaritalSex * ExtramaritalSex + MaritalStatus,  
       main = "Gender*Pre*Extra + MaritalStatus")
```



In the model [GPE][M], marital status depends in a complex way

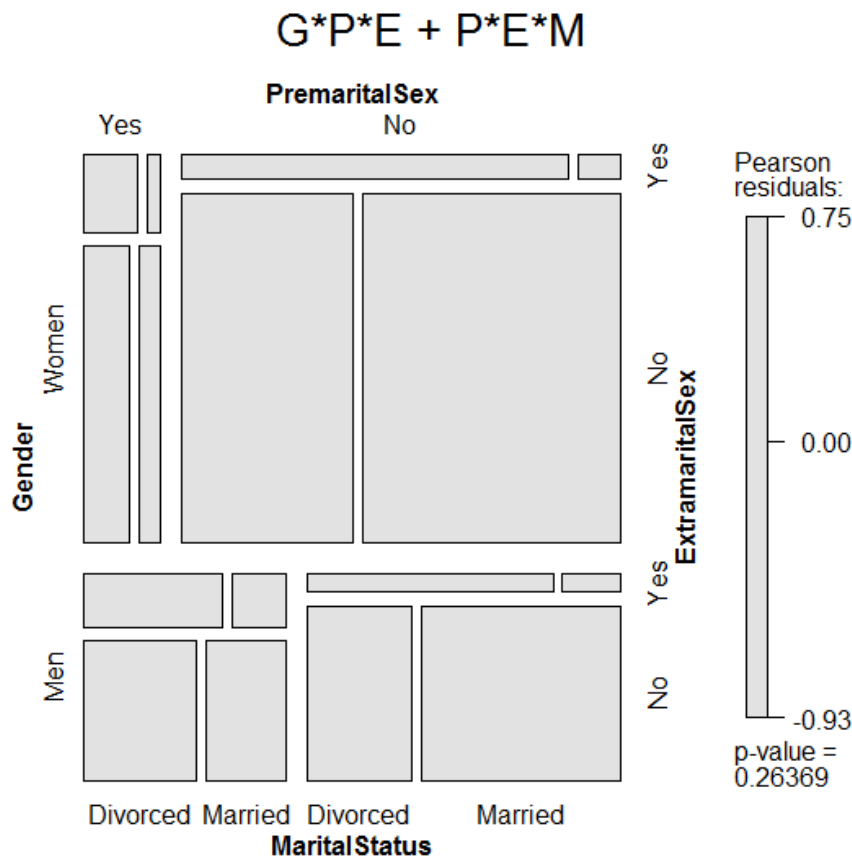
Among women, those reporting Pre- more likely to be divorced

Among men, those reporting Pre- only more likely to be divorced if Extra-

This suggests adding associations of M with P and E: [PEM] term

Mosaic plots: Full table

```
# (GPE) (PEM)
mosaic(PreSex,
      expected = ~ Gender * PremaritalSex * ExtramaritalSex
      + MaritalStatus * PremaritalSex * ExtramaritalSex,
      main = "G*P*E + P*E*M")
```



This model fits well, $G^2(4) = 5.26$, $p=0.26$

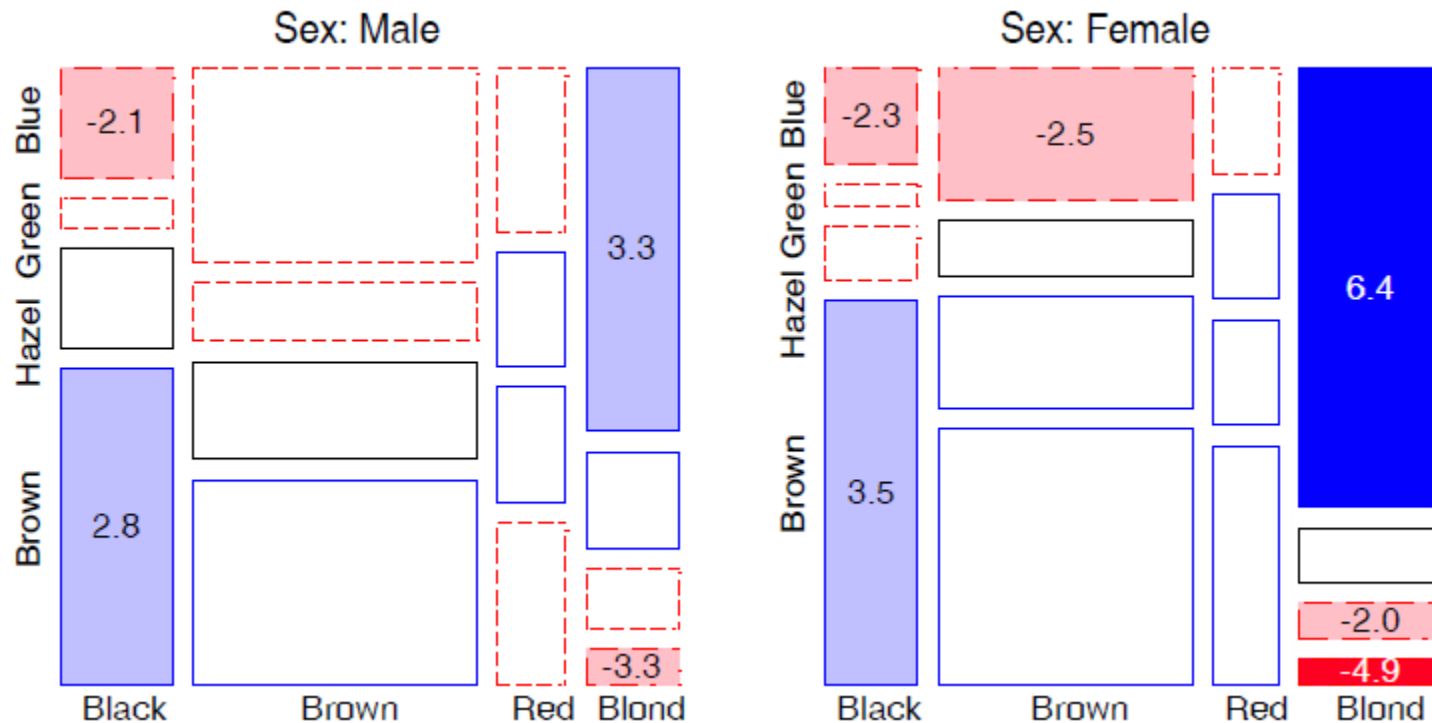
Loglinear thinking: once we take GPE into account, are there simpler models for association with M?

Looking forward: logit models for MaritalStatus often provide an easier path

Partial association, partial mosaics

Sometimes useful to do a [stratified analysis](#)

- How does association between two (or more) variables vary over levels of other variables?
- Mosaic plots for main variables show [partial association](#) at each level of others
- E.g., Hair color, Eye color, subset by Sex



Partial association, partial mosaics

Stratified analysis: conditional decomposition of G^2

- Fit models of partial (conditional) independence, $A \perp B \mid C_k$ at each level of (controlling for) C .
- $\Rightarrow$ partial G^2 's add to the overall G^2 for conditional independence, $A \perp B \mid C$

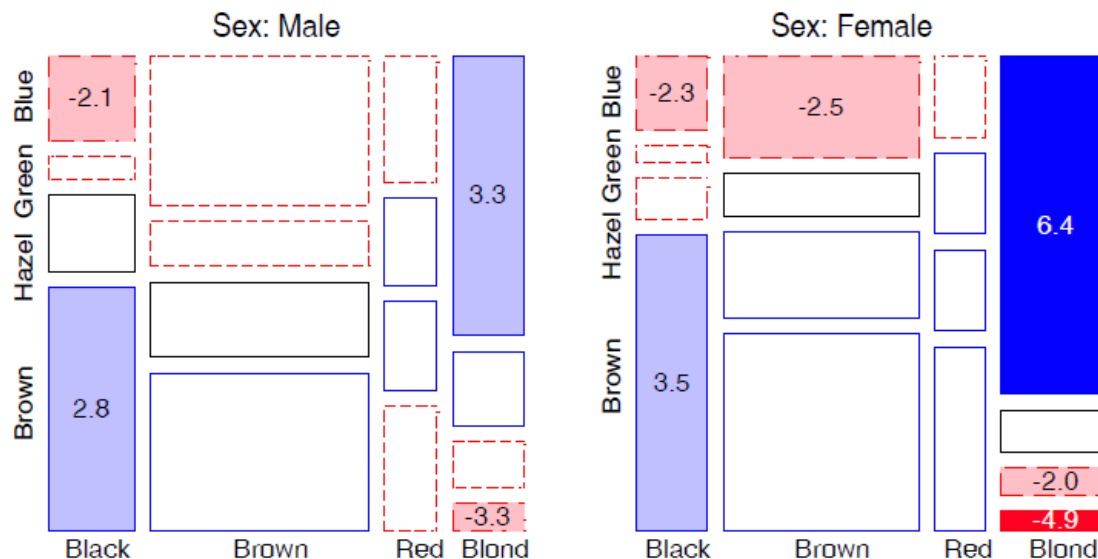
$$G^2_{A \perp B \mid C} = \sum_k G^2_{A \perp B \mid C(k)}$$

Table: Partial and Overall conditional tests, $Hair \perp Eye \mid Sex$

Model	df	G^2	p -value
$[Hair][Eye] \mid \text{Male}$	9	44.445	0.000
$[Hair][Eye] \mid \text{Female}$	9	112.233	0.000
$[Hair][Eye] \mid \text{Sex}$	18	156.668	0.000

Partial association: Summary

- Overall, there is a strong association of hair color and eye color, controlling for sex, $G^2_{(18)} = 156.67$
 - For F, $G^2_{(9)} = 112.23$ accounts for 72% of this association
- The pattern of association is similar for M & F
 - The largest difference is for blue-eyed blonds, much more prevalent among F than M. Is there a hair dye effect?



Summary: What we've learned

- Mosaic plots use sequential splits to show marginal and conditional frequencies in an n -way table
 - Shading: **sign** and **magnitude** of residuals $\rightarrow$ contributions to χ^2
 - Shows the pattern of association not accounted for
 - Permuting rows/cols often helps
- Loglinear models
 - Express associations with ANOVA-like interaction terms: $A*B, A*C$
 - Joint independence: $[AB][C] \equiv A * B + C$
 - Conditional independence: $[AC][BC] \equiv A \perp B \mid C$
 - Fitting models $\cong$ “cleaning the mosaic”
 - Response models: include all associations among predictors
- Sequential / partial plots & models
 - Sequential: Decompose all associations: $V_1; V_2|V_1; V_3|\{V_1, V_2\}, \dots$
 - Partial: Decompose conditional associations: $[V_1, V_2] \mid V_3 = \{a, b, \dots\}$